Introduction to Biostatistics

208141

Parichart Pattarapanitchai

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Preface

This book is an introduction to biostatistics for first-year health science students. It covers data types, summarization, probability distributions, estimation, hypothesis testing, and an introduction to statistical modeling.

All examples use health and clinical contexts.

Data in Health Science

Introduction

Why Does This Matter for Nurses?

Every day, nurses make decisions based on data — patient vital signs, lab results, medication dosages, and clinical outcomes. Understanding **what data is** and **how to classify it correctly** is the very first step toward making safe, evidence-based decisions in clinical practice.

In this chapter, we lay the foundation for everything that follows in this course.

Data Types

What is Data?

Data are pieces of information collected from observations or measurements. In health science, data come from patients, experiments, surveys, or medical records.

> **Example:** A nurse records the blood pressure, age, gender, pain score, and discharge status of every patient on the ward. Each of these recorded pieces of information is *data*.

Data can be broadly divided into two major types:

- **Categorical Data** — data that represent categories or groups
- **Numerical Data** — data that represent measurable quantities

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Categorical Data

Categorical data place individuals into groups. There are two subtypes:

Nominal Data

Nominal data have **no natural order** among categories. You can only say things are *different*, not *bigger* or *smaller*.

Nominal Examples in Nursing

- Blood type: A, B, AB, O
- Gender: Male, Female
- Disease diagnosis: Diabetes, Hypertension, Asthma
- Ward type: Medical, Surgical, Pediatric, ICU

Ordinal Data

Ordinal data have **a natural order**, but the *distance* between categories is not necessarily equal.

Ordinal Examples in Nursing

- Pain scale: None (0), Mild (1–3), Moderate (4–6), Severe (7–10)
- Patient satisfaction: Very dissatisfied Dissatisfied Neutral Satisfied Very satisfied
- Level of consciousness: Alert Confused Drowsy Unresponsive
- Education level: Primary Secondary Bachelor Graduate

Numerical Data

Numerical data represent actual measured or counted quantities. There are two subtypes:

Discrete Data

Discrete data are **counted** and can only take **whole number** values. There are no meaningful values between two consecutive counts.

Discrete Examples in Nursing

- Number of patients admitted per day: 0, 1, 2, 3, …
- Number of medication errors per month
- Number of children a patient has
- Number of hospital readmissions

Continuous Data

Continuous data are **measured** and can take **any value** within a range, including decimals.

Continuous Examples in Nursing

- Body temperature: 36.5°C, 37.2°C, 38.8°C
- Blood pressure: 120/80 mmHg, 135/85 mmHg
- Body weight: 58.3 kg, 72.5 kg
- Blood glucose level: 95.4 mg/dL

Summary: Data Type Classification

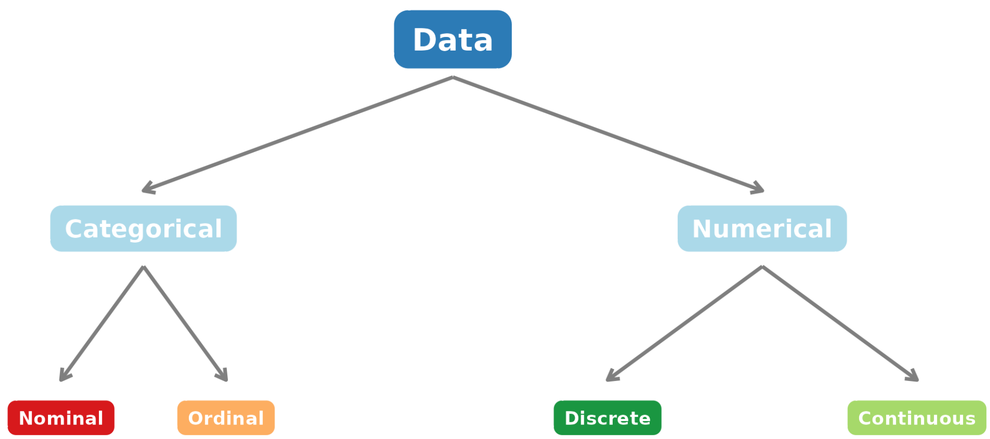

Figure 1: Classification of Data Types in Health Science

Population and Sample

Population

A **population** is the **complete set** of all individuals or observations that we are interested in studying.

In health research, populations are usually very large and it is **impractical or impossible** to collect data from everyone.

> **Example:** All patients with Type 2 diabetes in Thailand.

Sample

A **sample** is a **subset** of the population that is actually studied. We use the sample to draw conclusions (*make inferences*) about the whole population.

> **Example:** 200 patients with Type 2 diabetes selected from 5 hospitals in Bangkok.

> Key Principle
>
> A good sample must be **representative** of the population — meaning it should reflect the characteristics of the whole group, not just one part of it.

Key Terms

Table 1: Important Statistical Terms

Term	Definition	H
Population	The entire group of interest	A
Sample	A selected subset of the population	5

Parameter	A numerical summary describing the population (e.g., population mean mu)	T
Statistic	A numerical summary calculated from the sample (e.g., sample mean xbar)	T
Variable	A characteristic being measured or observed (e.g., age, blood pressure)	A
Observation	A single recorded value for one individual	N

Population vs. Sample: Visual Concept

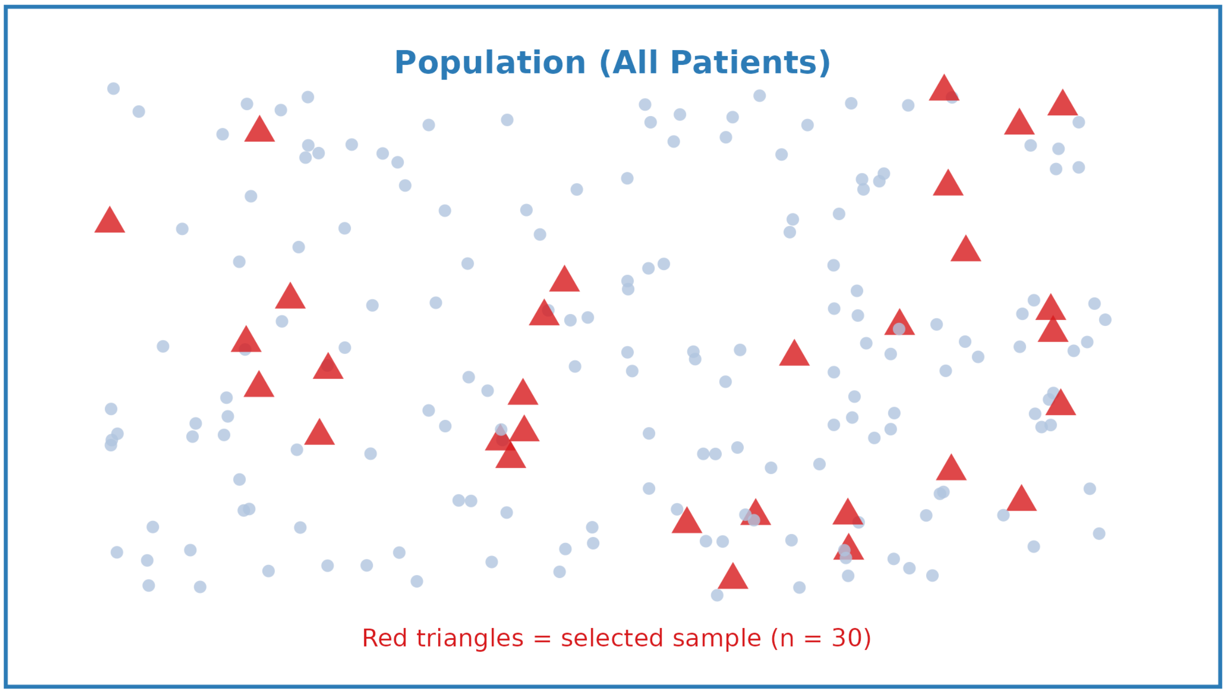

Figure 1: The sample is drawn from the population and used to make inferences

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Key Takeaways

Chapter 1 Summary

1. **Data** are recorded pieces of information collected from patients or research subjects.
2. Data are classified into two main types:
   - **Categorical**: Nominal (no order) and Ordinal (has order)
   - **Numerical**: Discrete (counted, whole numbers) and Continuous (measured, any value)
3. Identifying the **correct data type** is critical — it determines which statistical methods are appropriate.
4. A **population** is the entire group of interest; a **sample** is the subset we actually study.
5. We use **statistics** calculated from samples to estimate **parameters** of the population — this is the core idea behind all of biostatistics.

Examples

Example 1: Classify Each Variable

A nurse researcher collects the following information from 50 patients admitted to a medical ward:

Variable	Recorded Values
Patient ID	001, 002, 003, …
Age	45, 62, 38, … (years)
Gender	Male, Female
Blood type	A, B, AB, O
Systolic blood pressure	118, 135, 142, … (mmHg)
Pain score	0–10 scale
Number of medications	1, 2, 3, …
Discharge status	Recovered, Transferred, Deceased

Solution

Variable	Data Type	Reason
Patient ID	— (Identifier, not analyzed)	Just a label
Age	**Numerical – Continuous**	Measured, can be 45.5 years
Gender	**Categorical – Nominal**	Two categories, no order
Blood type	**Categorical – Nominal**	Four categories, no order
Systolic BP	**Numerical – Continuous**	Measured in mmHg, any value
Pain score	**Categorical – Ordinal**	Has order ($0 < 1 < \ldots < 10$), but gaps unequal
No. of medications	**Numerical – Discrete**	Counted, whole numbers only
Discharge status	**Categorical – Nominal**	Three categories, no natural order

Example 2: Population or Sample?

For each situation below, identify whether the group described is a **population** or a **sample**:

(a) A hospital wants to know the average job satisfaction of all 320 nurses working there. The HR department surveys all 320 nurses.

(b) A researcher wants to study pain levels in post-operative patients across Thailand. She recruits 150 patients from 3 hospitals in Chiang Mai.

(c) A nursing student studies the blood glucose levels of the 8 diabetic patients currently assigned to her ward.

Solution

(a) This is the **Population** — all 320 nurses are surveyed; no one is left out. The result is a *parameter*.
(b) This is a **Sample** — 150 patients from 3 hospitals cannot represent all post-operative patients in Thailand. The result is a *statistic* used to estimate the population parameter.
(c) This is a **Sample** — the 8 patients on the ward are a small subset of all diabetic patients.

Exercises

Exercise 1

A public health nurse collects the following data from patients visiting a community health clinic:

- Weight (kg)
- Presence of hypertension (Yes / No)
- Severity of hypertension (Mild / Moderate / Severe)
- Number of clinic visits in the past year
- Ethnicity
- Body temperature (°C)
- Diastolic blood pressure (mmHg)

Classify each variable into the correct data type and explain your reasoning.

Exercise 2

A researcher is studying the prevalence of depression among elderly patients in nursing homes across the country. She selects 200 patients from 4 nursing homes in the central region.
(a) What is the population in this study?
(b) What is the sample?
(c) If the average depression score of the 200 patients is 14.2, is this number a *parameter* or a *statistic*? Explain.

Exercise 3

Look at the following list and decide whether each is an example of a **discrete** or **continuous** variable:

1. The number of patients who fell during a hospital shift
2. The time (in minutes) a patient waited before being seen by a doctor
3. The number of syringes used during a procedure
4. A patient's oxygen saturation level (SpO %)
5. The number of newborns delivered in a hospital per week

End of Chapter 1

Coming Up Next

Chapter 2: Data Summarization — How do we make sense of a large set of patient data? We will learn to calculate means, medians, standard deviations, and draw charts that tell a clear story.

Data Summarization

Introduction

Why Does This Matter for Nurses?

Imagine you have blood pressure readings from 80 patients on your ward. Looking at 80 individual numbers tells you very little. But if someone says *"the average systolic BP is 138 mmHg and most patients fall between 120 and 155 mmHg"* — now you have a clear picture.

Data summarization is the process of turning raw data into meaningful, easy-to-understand information using numbers and charts. This is one of the most practical skills in clinical nursing and health research.

Numerical Summarization

Numerical summarization uses **numbers** to describe the key features of data. There are two important aspects we want to describe:

1. **Central tendency** — Where is the "middle" or "typical" value?
2. **Variability** — How spread out are the values?

Measures of Central Tendency

Mean (Arithmetic Average)

The **mean** is calculated by adding all values and dividing by the number of observations.

$$\bar{x} = \frac{\sum_{i=1}^{n} x_i}{n} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

Where:

- $\bar{x}$ = sample mean (read as "x-bar")
- x_i = each individual value
- n = total number of observations
- $\sum$ = "sum of all"

> When to Use the Mean
>
> The mean works best for **continuous or discrete numerical data** that is **roughly symmetric** (not heavily skewed) and has **no extreme outliers**. A single extreme value can pull the mean far from the typical value.

Median

The **median** is the **middle value** when all observations are arranged in order.

- If n is **odd**: the median is the value at position $\frac{n+1}{2}$
- If n is **even**: the median is the **average of the two middle values** at positions $\frac{n}{2}$ and $\frac{n}{2}+1$

> When to Use the Median
>
> The median is preferred when data is **skewed** or has **outliers**, because it is not affected by extreme values. It is also the appropriate measure for **ordinal data**.

Mode

The **mode** is the value that appears **most frequently** in the data.

- A dataset can have **one mode** (unimodal), **two modes** (bimodal), or **no mode** (all values appear equally).
- The mode is the only measure of central tendency appropriate for **nominal data**.

Choosing the Right Measure

Table 1: Choosing the Right Measure of Central Tendency

Measure	Best For	Nursing Example
Mean	Numerical data (continuous or discrete), symmetric distribution	Average patient age
Median	Numerical data with skewness or outliers; Ordinal data	Average length of h
Mode	Nominal data; finding the most common category	Most common bloo

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Measures of Variability

Knowing the average alone is not enough. Two wards might have the same average patient age but very different spreads — one ward may have all patients aged 55–65, another may have patients aged 20–90. Variability tells us how **spread out** the data are.

Range

The **range** is the simplest measure of spread:

$$\text{Range} = \text{Maximum value} - \text{Minimum value}$$

> Limitation
>
> The range only uses two values (max and min) and is very sensitive to outliers.

Variance

The **variance** measures the average squared distance of each value from the mean.

Population variance:

$$\sigma^2 = \frac{\sum_{i=1}^{N}(x_i - \mu)^2}{N}$$

Sample variance (used more often in practice):

$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}$$

Where:

- μ = population mean, $\bar{x}$ = sample mean
- N = population size, n = sample size
- $n - 1$ in the denominator corrects for the fact that we're using a sample

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Standard Deviation

The **standard deviation (SD)** is the square root of the variance. It is expressed in the **same units as the original data**, making it much easier to interpret than variance.

$$s = \sqrt{s^2} = \sqrt{\frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1}}$$

Interpreting Standard Deviation

A **small SD** means values are clustered closely around the mean. A **large SD** means values are widely spread.
In clinical practice, you may see lab reports written as: **Mean ± SD**, for example: Hemoglobin = **12.5 ± 1.8 g/dL**

Summary: Numerical Measures

Table 2: Summary of Numerical Summary Measures

Measure	Symbol	Purpose	Appropriate Data Type
Mean	xbar	Central tendency	Numerical
Median	Md	Central tendency	Numerical / Ordinal
Mode	Mo	Central tendency	Any
Range	Max - Min	Spread	Numerical
Variance	s2	Spread (squared units)	Numerical
Standard Deviation	s	Spread (original units)	Numerical

Pictorial Summarization

Charts and graphs turn numbers into visual stories. The correct chart depends on the **data type**.

Bar Chart

Used for **categorical data** (nominal or ordinal). Each bar represents a category; the height shows the count or percentage.

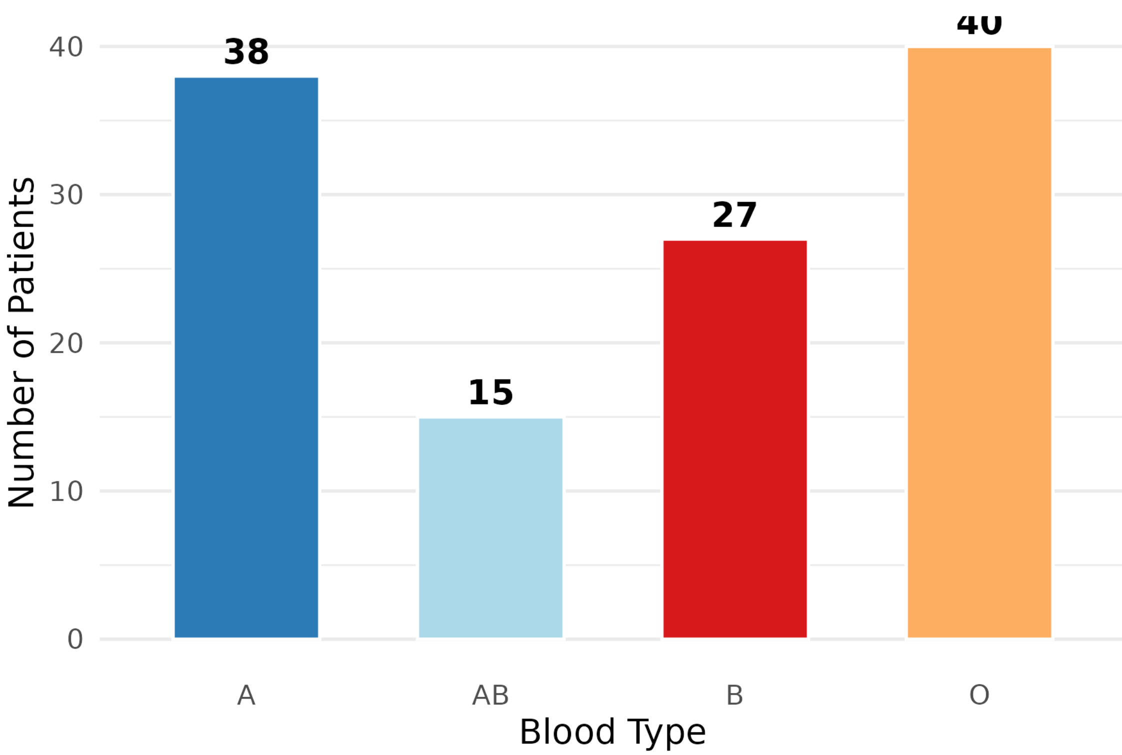

Figure 1: Bar Chart: Blood Types of 120 Patients in a Medical Ward

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Histogram

Used for **continuous numerical data**. Bars are drawn over class intervals (ranges of values); bars touch each other because the data is continuous.

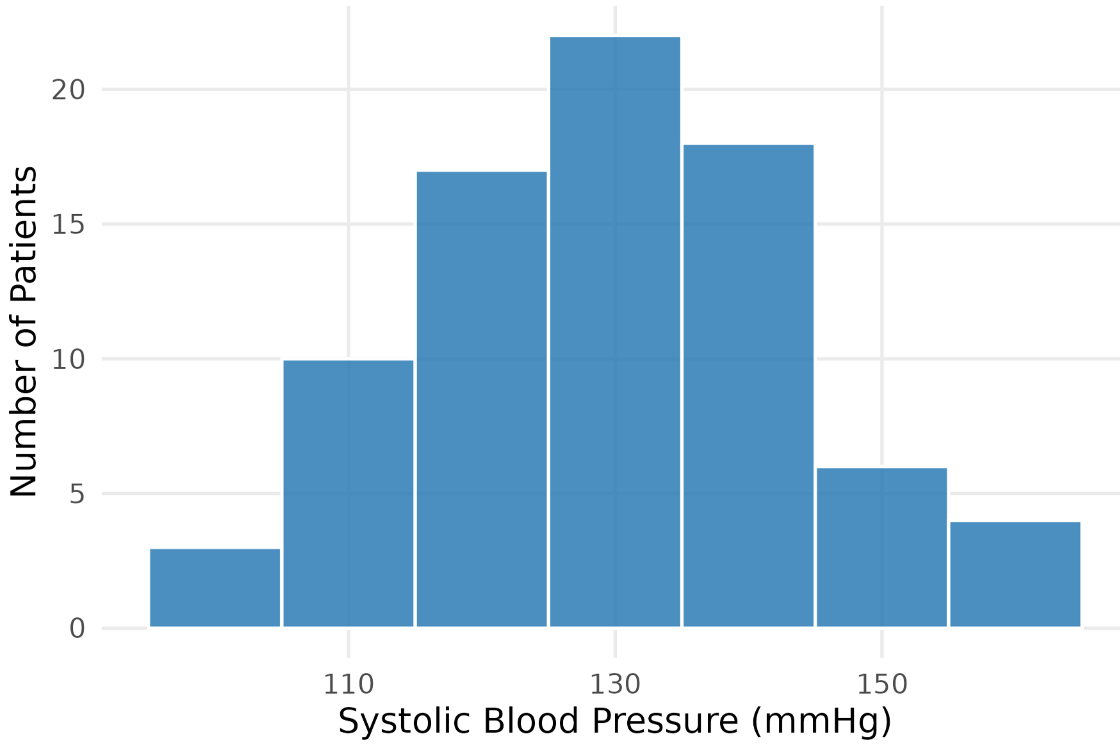

Figure 2: Histogram: Systolic Blood Pressure of 80 Patients

Bar Chart vs Histogram

	Bar Chart	Histogram
Data type	Categorical	Continuous numerical
Bars	Separated (gaps)	Touching (no gaps)
X-axis	Categories	Numerical ranges

Box Plot (Box-and-Whisker Plot)

A **box plot** summarizes the **distribution** of numerical data using 5 key values (the **five-number summary**):

Value	Description
Minimum	Smallest value (excluding outliers)
Q1	25th percentile (lower quartile)
Median (Q2)	50th percentile — middle value
Q3	75th percentile (upper quartile)
Maximum	Largest value (excluding outliers)

The **Interquartile Range (IQR) = Q3 − Q1** represents the middle 50% of data. Points beyond **1.5 × IQR** from Q1 or Q3 are marked as **outliers ()**.

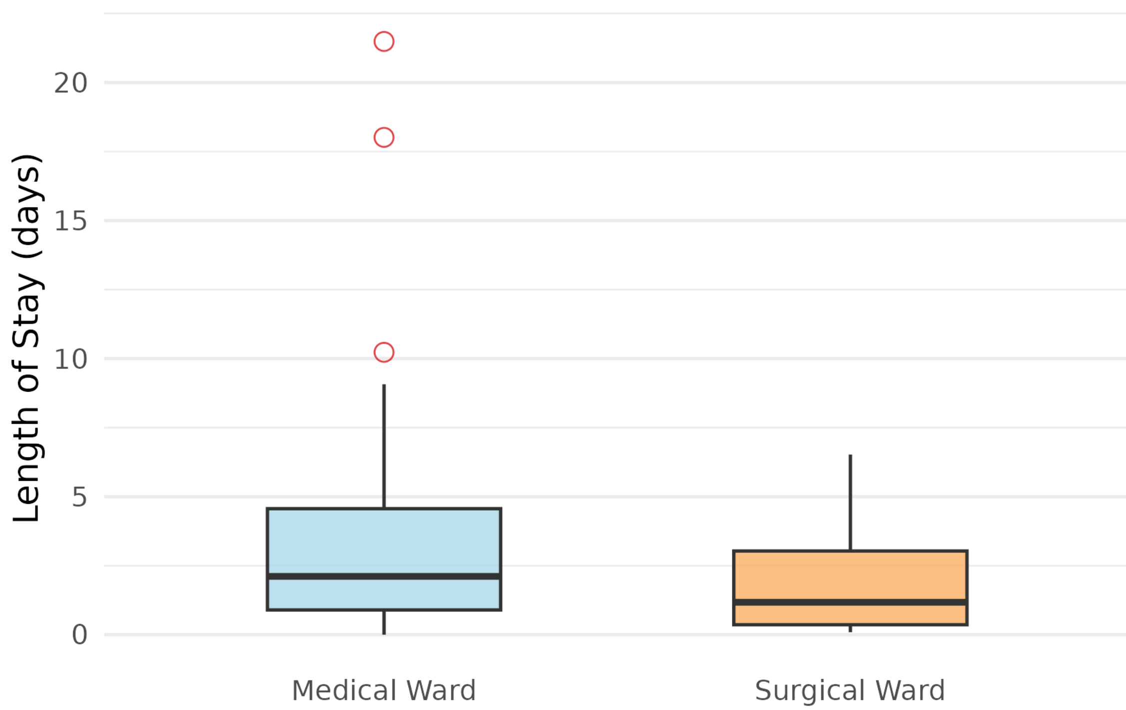

Figure 3: Box Plot: Comparing Length of Hospital Stay Between Two Wards

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Choosing the Right Chart

Table 1: Guide to Choosing the Right Chart

Chart Type	Data Type	What It Shows

Bar Chart	Categorical (Nominal/Ordinal)	Frequency or percentage of each category
Pie Chart	Categorical (Nominal) - proportions	Proportion of each category as a slice of the
Histogram	Numerical (Continuous)	Distribution shape, spread, and central tend
Box Plot	Numerical (Continuous/Discrete)	Median, spread, quartiles, and outliers

Key Takeaways

Chapter 2 Summary

1. **Numerical summarization** uses measures of *central tendency* (mean, median, mode) and *variability* (range, variance, SD) to describe data with numbers.
2. The **mean** is best for symmetric numerical data; the **median** is better when data is skewed or has outliers; the **mode** is used for categorical data.
3. The **standard deviation (SD)** is the most commonly reported measure of spread in clinical research, expressed in the same units as the data.
4. **Pictorial summarization** uses charts: bar charts for categorical data, histograms for continuous data, and box plots to compare groups and identify outliers.
5. Always choose the **correct summary measure and chart** based on the **data type** — using the wrong one leads to misleading conclusions.

Examples

Example 1: Calculating Mean, Median, and SD

A nurse records the resting heart rate (beats per minute) of **9 patients**:

$$72, \ 68, \ 85, \ 74, \ 90, \ 68, \ 78, \ 72, \ 110$$

Calculate the **(a) mean**, **(b) median**, and **(c) standard deviation**.

Solution

Step 1 — Arrange data in order:

$$68, \ 68, \ 72, \ 72, \ 74, \ 78, \ 85, \ 90, \ 110$$

(a) Mean:

$$\bar{x} = \frac{68 + 68 + 72 + 72 + 74 + 78 + 85 + 90 + 110}{9} = \frac{717}{9} = 79.7 \text{ bpm}$$

(b) Median:
$n = 9$ (odd), so median is at position $\frac{9+1}{2} = 5\text{th}$ value.
Counting from the ordered list: the 5th value = **74 bpm**
Notice that the median (74) is lower than the mean (79.7) because the outlier value of 110 pulls the mean upward.

(c) Standard Deviation:
First, find each $(x_i - \bar{x})^2$:

Table 1: Step-by-step SD calculation

Patient	Heart Rate (x)	x - xbar	(x - xbar)2

Patient 1	68	-11.7	136.11
Patient 2	68	-11.7	136.11
Patient 3	72	-7.7	58.78
Patient 4	72	-7.7	58.78
Patient 5	74	-5.7	32.11
Patient 6	78	-1.7	2.78
Patient 7	85	5.3	28.44
Patient 8	90	10.3	106.78
Patient 9	**110**	**30.3**	**920.11**

$$\sum (x_i - \bar{x})^2 = 1480$$

$$s^2 = \frac{1480}{9-1} = \frac{1480}{8} = 185$$

$$s = \sqrt{185} = 13.6 \text{ bpm}$$

Interpretation: The average heart rate is **79.7 ± 13.5 bpm**. The relatively large SD reflects the wide spread, largely influenced by the patient with 110 bpm.

Example 2: Choosing the Right Summary and Chart

A hospital administrator records the **diagnosis** of 200 patients admitted last month and wants to summarize the data. A research nurse records the **length of stay (days)** for the same 200 patients.

Solution

Variable	Data Type	Best Summary Measure	Best Chart
Diagnosis	Categorical – Nominal	Mode (most common diagnosis)	Bar Chart
Length of stay	Numerical – Continuous (likely skewed)	Median + IQR	Box Plot or Histogram

Why median for length of stay? Hospital stays are almost always right-skewed — most patients stay a few days, but a small number may stay for weeks or months. These long-stay patients would pull the mean upward, making it misleading. The median better represents the "typical" patient.

Exercises

Exercise 1

A nurse records the body temperature (°C) of **10 patients** at 8:00 AM:

$$37.2,\ 36.8,\ 38.5,\ 37.0,\ 39.1,\ 36.8,\ 37.5,\ 38.0,\ 37.2,\ 40.2$$

(a) Calculate the **mean** body temperature.
(b) Find the **median** body temperature.
(c) What is the **mode**?
(d) Calculate the **range**.
(e) Calculate the **standard deviation**. Show all steps.
(f) One patient has a temperature of 40.2°C. How does this value affect the mean compared to the median? Which measure would you report to the physician, and why?

Exercise 2

A community health nurse surveys **50 elderly patients** and records:

- **Gender** (Male / Female)
- **Systolic blood pressure** (mmHg)
- **Self-rated health** (Poor / Fair / Good / Very Good / Excellent)

For each variable, state:
(a) The data type.
(b) The most appropriate measure of central tendency.
(c) The most appropriate chart to display the data.

Exercise 3

The number of patient falls recorded each month over **12 months** in a hospital ward are:

$$2,\ 0,\ 3,\ 1,\ 4,\ 2,\ 1,\ 0,\ 3,\ 2,\ 5,\ 1$$

(a) Calculate the mean and median number of falls per month.
(b) Draw a **bar chart** (by hand or describe what it would look like) showing the frequency of each number of falls.
(c) The ward manager wants to set a monthly target. Should she use the mean or

median as the baseline? Explain.

End of Chapter 2

Coming Up Next

Chapter 3: Random Variable and Probability Distribution — We will learn what a random variable is, how probabilities work, and why the **normal distribution** (bell curve) is so important in health science research.

Random Variable and Probability Distribution

Introduction

Why Does This Matter?

When we collect data — patient ages, blood pressure readings, test results — we rarely get the same value twice. Outcomes vary, and that variation follows patterns we can describe mathematically.
Understanding **how data is distributed** allows us to answer questions like:

- What is the probability that a randomly selected patient has a fever above 39°C?
- What proportion of adults have a BMI in the healthy range?
- How unusual is a blood pressure reading of 160 mmHg in this population?

In this chapter, we introduce the concepts of **frequency distributions** and the **normal distribution** — the most important probability distribution in all of biostatistics.

Frequency Distributions

What Is a Frequency Distribution?

A **frequency distribution** is a systematic way of organizing raw data into a table that shows how often each value (or range of values) occurs.

It answers the simple question: *"How many times does each value appear?"*

Frequency Distribution for Categorical Data

For categorical variables, we simply count how many observations fall into each category.

Example

Blood types of 200 patients admitted to a hospital last month:

Table 1: Frequency Distribution: Blood Type of 200 Patients

Blood Type	Frequency (f)	Relative Frequency (f/n)	Percentage
A	76	$76/200 = 0.38$	38%
B	54	$54/200 = 0.27$	27%
AB	30	$30/200 = 0.15$	15%
O	40	$40/200 = 0.20$	20%
Total	**200**	$\mathbf{200/200 = 1.00}$	**100%**

Relative frequency $= \dfrac{f}{n}$, where f is the count for that category and n is the total number of observations.

Frequency Distribution for Continuous Data

For continuous variables, we group values into **class intervals** (also called *bins*) and count how many observations fall in each interval.

Steps to construct a frequency distribution table:

1. Find the **range** = Maximum − Minimum
2. Choose the **number of classes** (typically 5–10)
3. Calculate **class width** = $\frac{\text{Range}}{\text{Number of classes}}$ (round up)
4. List the class intervals and count observations in each

Example

Systolic blood pressure (mmHg) of 80 patients:

Table 2: Frequency Distribution: Systolic Blood Pressure of 80 Patients

Class Interval (mmHg)	Frequency (f)	Relative Frequency	Cumulative Frequency
95-104	2	0.025	2
105-114	9	0.112	11
115-124	17	0.212	28
125-134	22	0.275	50
135-144	18	0.225	68
145-154	8	0.100	76
155-164	4	0.050	80
165-174	0	0.000	80

Cumulative frequency tells us how many observations fall *at or below* the upper boundary of each class — useful for answering questions like *"What proportion of patients have SBP below 140 mmHg?"*

Visualizing Frequency Distributions

A frequency distribution table can be visualized as a **histogram**, where the height of each bar represents the frequency of that class interval.

The **shape** of the histogram is important — it tells us about the **distribution** of the data, which leads us to the next section.

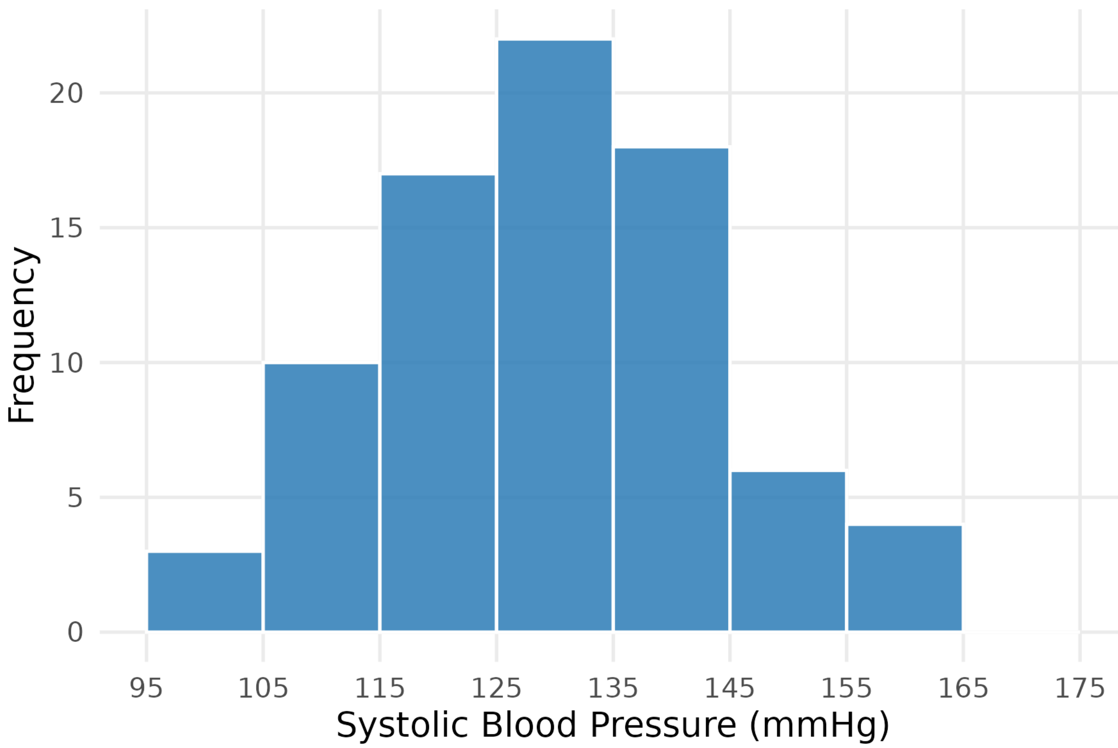

Figure 1: Histogram of Systolic Blood Pressure (80 patients)

The Normal Distribution

What Is the Normal Distribution?

The **normal distribution** (also called the *Gaussian distribution* or *bell curve*) is the most important probability distribution in statistics.

Many biological and clinical measurements naturally follow — or approximately follow — a normal distribution:

- Adult height and body weight
- Blood pressure in a healthy population
- Serum cholesterol levels
- Birth weight of newborns

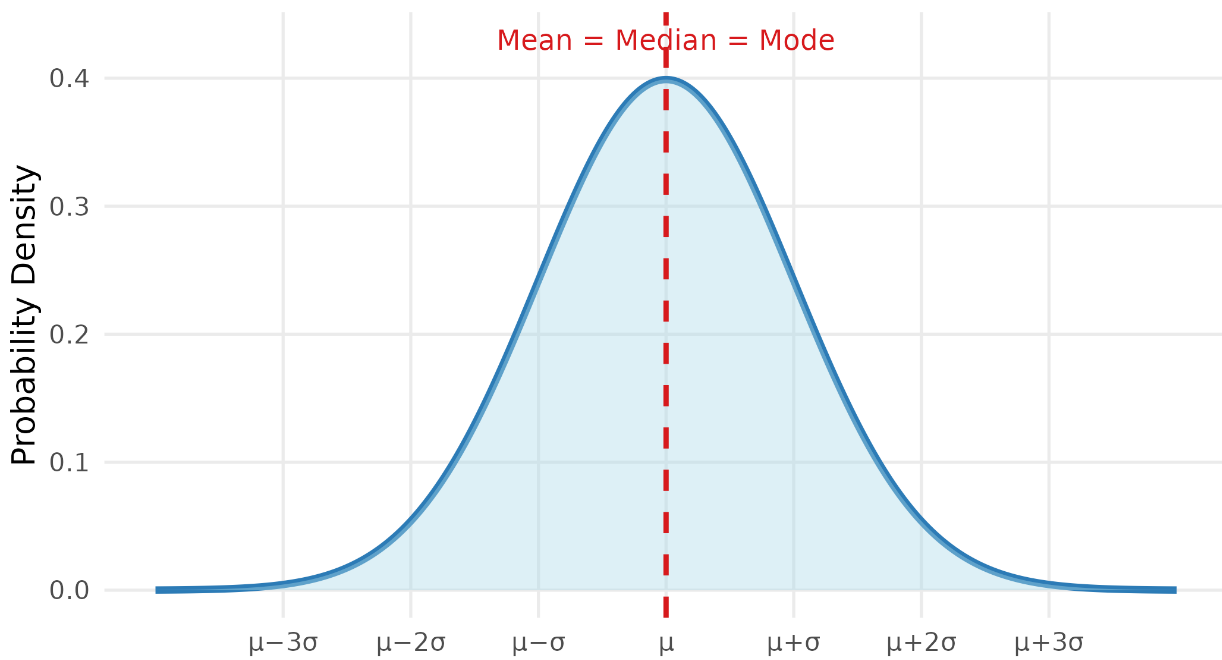

Figure 1: The Normal Distribution: a symmetric, bell-shaped curve

Key Properties of the Normal Distribution

A normal distribution is completely described by just **two parameters**:

Parameter	Symbol	Meaning
Mean	μ	Center of the distribution (peak of the bell)
Standard deviation	σ	Width of the bell — larger = wider, flatter curve

Key properties:

1. **Symmetric** around the mean — the left and right halves are mirror images
2. **Mean = Median = Mode** — all three are equal and located at the center
3. The **total area** under the curve = 1 (representing 100% of all probability)
4. The curve extends infinitely in both directions but never touches the x-axis

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The 68–95–99.7 Rule (Empirical Rule)

One of the most useful facts about the normal distribution is how much of the data falls within 1, 2, or 3 standard deviations of the mean.

The Empirical Rule

Range	Proportion of data
Within **1 SD** of the mean: $\mu \pm \sigma$	**68%**
Within **2 SD** of the mean: $\mu \pm 2\sigma$	**95%**
Within **3 SD** of the mean: $\mu \pm 3\sigma$	**99.7%**

This means only about **5%** of values fall more than 2 SD from the mean, and only **0.3%** fall more than 3 SD away — these are considered unusual or extreme.

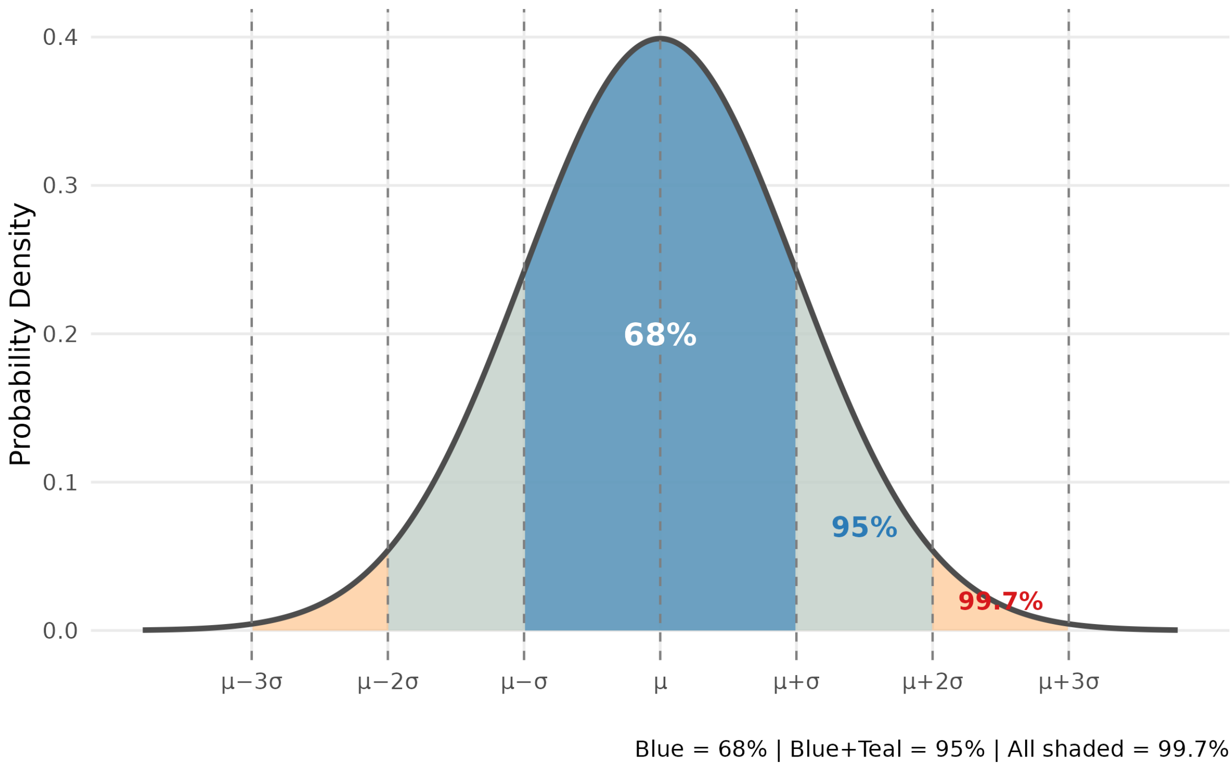

Figure 2: The 68–95–99.7 Rule for the Normal Distribution

The Standard Normal Distribution and Z-Score

The Z-Score

Different normal distributions have different means and SDs. To compare values across different distributions, we **standardize** them using the **Z-score**:

$$z = \frac{x - \mu}{\sigma}$$

Where:

- x = the observed value
- μ = the population mean
- σ = the population standard deviation
- z = the number of standard deviations x is away from the mean

A **positive Z** means the value is above the mean; a **negative Z** means it is below the mean.

The Standard Normal Distribution

When we convert all values to Z-scores, the resulting distribution is called the **standard normal distribution**, with:

$$\mu = 0 \quad \text{and} \quad \sigma = 1$$

This is written as $Z \sim N(0, 1)$.

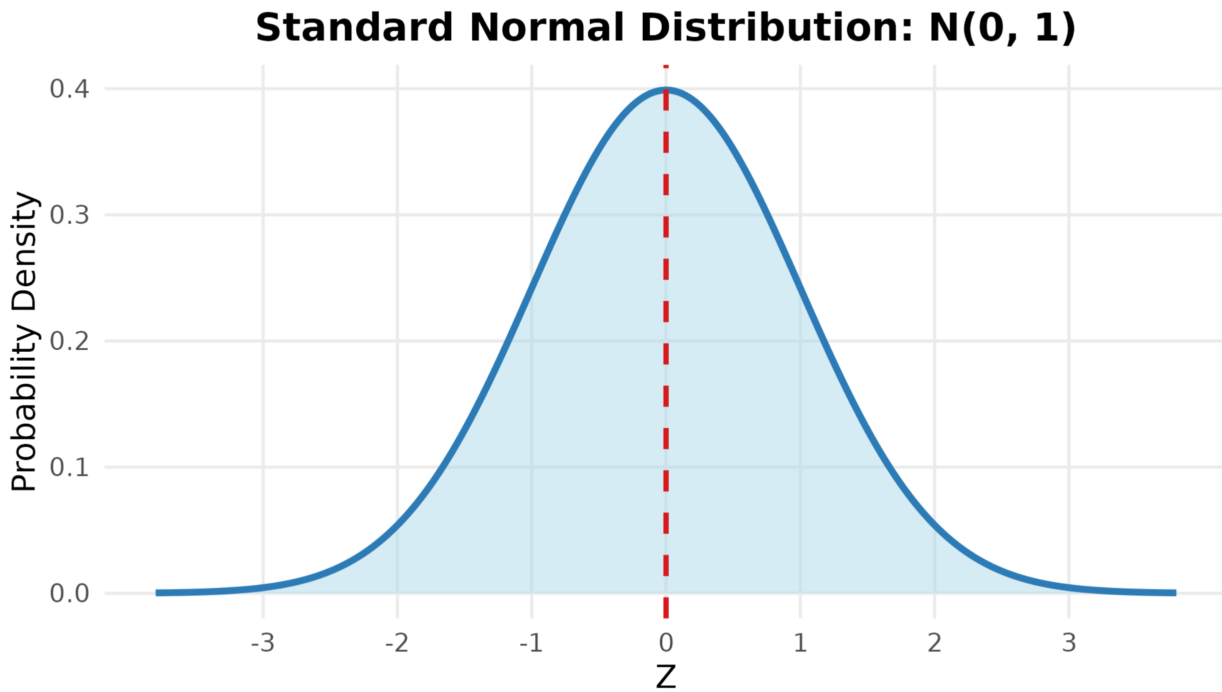

Figure 3: Standard Normal Distribution N(0,1)

Finding Probabilities Using the Z-Table

The **Z-table** (also called the standard normal table) gives us the **area to the left** of any Z-score — which equals the **probability** that a randomly selected observation falls below that value.

Table 1: Area to the left of Z (rows = Z integer + 1st decimal, columns = 2nd decimal)

	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09

0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817

How to use the table: To find $P(Z < 1.23)$, look at row **1.2** and column **0.03** read off the value.

Common Z-Table Probability Rules

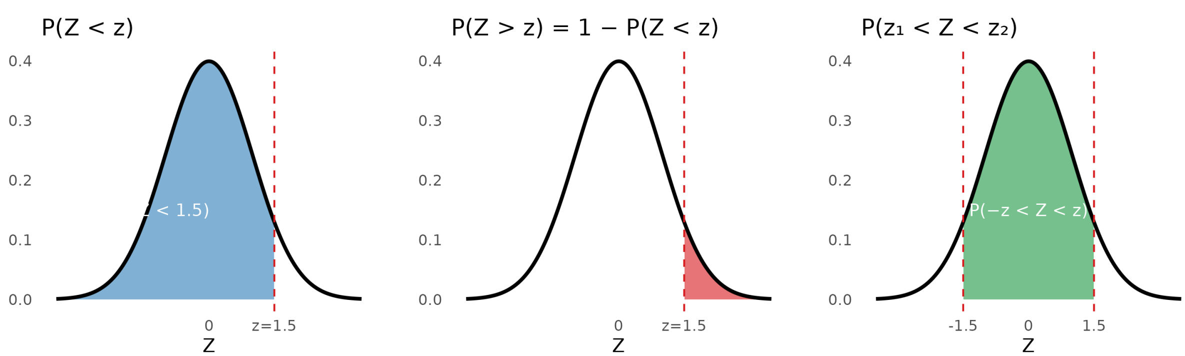

Figure 4: Three useful probability rules for the normal distribution

Key Probability Rules

Situation	Formula
Probability **below** z	$P(Z < z)$ — read directly from Z-table
Probability **above** z	$P(Z > z) = 1 - P(Z < z)$
Probability **between** two values	$P(z_1 < Z < z_2) = P(Z < z_2) - P(Z < z_1)$

Key Takeaways

Chapter 3 Summary

1. A **frequency distribution** organizes data into a table showing how often each value or range of values occurs. It can show absolute frequency, relative frequency, or cumulative frequency.

2. The **normal distribution** is a symmetric, bell-shaped distribution described by its mean (μ) and standard deviation (σ).

3. The **68–95–99.7 rule**: approximately 68%, 95%, and 99.7% of values fall within 1, 2, and 3 standard deviations of the mean, respectively.

4. A **Z-score** measures how many standard deviations a value is from the mean: $z = \dfrac{x - \mu}{\sigma}$

5. The **Z-table** gives the probability (area) to the left of any Z-score, allowing us to find the probability that a measurement falls in any range.

Examples

Example 1: Constructing a Frequency Distribution Table

The following are the ages (in years) of 20 patients:

$$45, 62, 38, 55, 71, 48, 60, 33, 52, 67,$$
$$41, 58, 74, 50, 65, 44, 57, 69, 36, 53$$

Construct a frequency distribution table using **5 class intervals**.

Solution

Step 1 — Find the range:

$$\text{Range} = 74 - 33 = 41$$

Step 2 — Calculate class width:

$$\text{Class width} = \frac{41}{5} = 8.2 \rightarrow \text{round up to } 9$$

Step 3 — Construct the table (starting from 33):

Table 1: Frequency Distribution of Patient Ages (n = 20)

Age Group (years)	Frequency (f)	Relative Frequency	Percentage	Cumulative Frequency
33-41	4	0.2	20%	4
42-50	4	0.2	20%	8
51-59	6	0.3	30%	14
60-68	4	0.2	20%	18
69-77	2	0.1	10%	20

Interpretation: The largest group of patients is aged **51–59 years** (6 patients, 30%). Half of all patients (cumulative frequency = 10) are aged 50 years or younger.

Example 2: Applying the Empirical Rule

The systolic blood pressure (SBP) of adults in a community follows a normal distribution with **mean = 125 mmHg** and **SD = 14 mmHg**.

(a) Between what two values does the SBP of the **middle 95%** of adults fall?

(b) What percentage of adults have SBP **above 153 mmHg**?

Solution

(a) The middle 95% falls within $\mu \pm 2\sigma$:

$$125 - 2(14) = 97 \text{ mmHg} \quad \text{to} \quad 125 + 2(14) = 153 \text{ mmHg}$$

So **95% of adults** have SBP between **97 and 153 mmHg**.
(b) SBP = 153 mmHg is exactly $\mu + 2\sigma$. Since 95% of values fall *within* 2 SD, the remaining 5% fall *outside*. By symmetry, **2.5%** fall above 153 mmHg and 2.5% fall below 97 mmHg.
Answer: Approximately **2.5%** of adults have SBP above 153 mmHg.

Example 3: Using Z-Scores and the Z-Table

Body weight of adult males in a region is approximately normally distributed with **mean = 70 kg** and **SD = 10 kg**.

(a) What is the probability that a randomly selected adult male weighs **less than 85 kg**?

(b) What is the probability that he weighs **more than 55 kg**?

(c) What is the probability that he weighs **between 60 and 80 kg**?

Solution

(a) P(X < 85):

$$z = \frac{85 - 70}{10} = \frac{15}{10} = 1.50$$

From Z-table: $P(Z < 1.50) = \mathbf{0.9332}$
93.3% of adult males weigh less than 85 kg.

(b) P(X > 55):

$$z = \frac{55 - 70}{10} = \frac{-15}{10} = -1.50$$

$$P(Z > -1.50) = 1 - P(Z < -1.50) = 1 - 0.0668 = \mathbf{0.9332}$$

93.3% of adult males weigh more than 55 kg. *(This makes sense by symmetry — 55 and 85 are equidistant from the mean.)*

(c) P(60 < X < 80):

$$z_1 = \frac{60 - 70}{10} = -1.00 \qquad z_2 = \frac{80 - 70}{10} = 1.00$$

$$P(-1.00 < Z < 1.00) = P(Z < 1.00) - P(Z < -1.00)$$
$$= 0.8413 - 0.1587 = \mathbf{0.6826}$$

68.3% of adult males weigh between 60 and 80 kg — consistent with the 68–95–99.7 rule.

Exercises

Exercise 1

The fasting blood glucose levels (mg/dL) of 25 patients are:

$$95, 112, 88, 134, 102, 119, 145, 97, 108, 125,$$

$$91, 130, 103, 116, 88, 142, 99, 121, 107, 138,$$

$$94, 115, 126, 100, 111$$

(a) Construct a frequency distribution table with **5 class intervals**. Include columns for frequency, relative frequency, percentage, and cumulative frequency.
(b) Which class interval contains the most patients?
(c) What percentage of patients have a blood glucose level below 120 mg/dL?

Exercise 2

The birth weight of full-term newborns is approximately normally distributed with **mean = 3,300 g** and **SD = 400 g**.
Using the **68–95–99.7 rule**:
(a) What range of birth weights includes the **middle 68%** of newborns?
(b) What range includes the **middle 99.7%** of newborns?
(c) A newborn weighing less than **2,500 g** is classified as low birth weight. Based on the empirical rule, is this an unusual outcome? Explain.

Exercise 3

The resting heart rate of healthy adults follows a normal distribution with **mean = 72 bpm** and **SD = 8 bpm**.
Find the following probabilities using the Z-table. Show the Z-score calculation and draw a sketch of the normal curve for each part.
(a) $P(\text{heart rate} < 80)$
(b) $P(\text{heart rate} > 88)$
(c) $P(64 < \text{heart rate} < 88)$
(d) A heart rate below 60 bpm is considered **bradycardia**. What proportion of healthy adults would be classified as having bradycardia based on this distribution?

End of Chapter 3

Coming Up Next

Chapter 4: Estimation — Using what we know about the normal distribution, we will learn how to estimate population parameters from sample data and construct **confidence intervals** that quantify our uncertainty.

Estimation

Introduction

Why Does This Matter?

In practice, we almost never have access to an entire population. We collect data from a **sample** and use it to draw conclusions about the **population** — a process called **statistical inference**.
Estimation is one of the two main branches of statistical inference (the other being hypothesis testing, covered in Chapter 5).
In this chapter, we answer questions like:

- The mean serum cholesterol in our sample is 198 mg/dL — what can we say about the mean for the *entire* population?
- A survey found that 34% of respondents have hypertension — what is a reasonable range for the true population proportion?

The key tool for answering these questions is the **confidence interval**.

Concept of Estimation

Point Estimate vs. Interval Estimate

There are two ways to estimate a population parameter from sample data:

Point Estimate

A **point estimate** is a **single value** calculated from the sample that serves as our best guess for the population parameter.

Population Parameter	Symbol	Point Estimate	Symbol
Population mean	μ	Sample mean	$\bar{x}$
Population proportion	p	Sample proportion	$\hat{p}$
Population SD	σ	Sample SD	s

A point estimate is simple and easy to communicate, but it gives us **no information about how accurate or reliable** that single number is.

Interval Estimate (Confidence Interval)

An **interval estimate** gives a **range of plausible values** for the population parameter, along with a stated level of confidence.

$$\text{Point Estimate} \pm \text{Margin of Error}$$

This range is called a **confidence interval (CI)**.

Key Concept: Confidence Level

The **confidence level** (commonly **90%, 95%, or 99%**) tells us:

"If we repeated this study many times and calculated a confidence interval

each time, approximately X% of those intervals would contain the true population parameter.”

In health research, **95% confidence** is the most widely used standard.

The Role of the Standard Error

Before constructing a confidence interval, we need to understand the **standard error (SE)** — a measure of how much the sample estimate varies from sample to sample.

Standard Error of the Mean

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}}$$

Where s is the sample standard deviation and n is the sample size.

Standard Error of a Proportion

$$SE_{\hat{p}} = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

SE vs. SD — What Is the Difference?

	Standard Deviation (SD)	Standard Error (SE)
Describes	Variability of individual observations	Variability of the sample mean
Gets smaller with	More homogeneous data	Larger sample size
Used for	Describing the spread of data	Constructing confidence intervals

The SE always gets smaller as n increases — larger samples give more precise estimates.

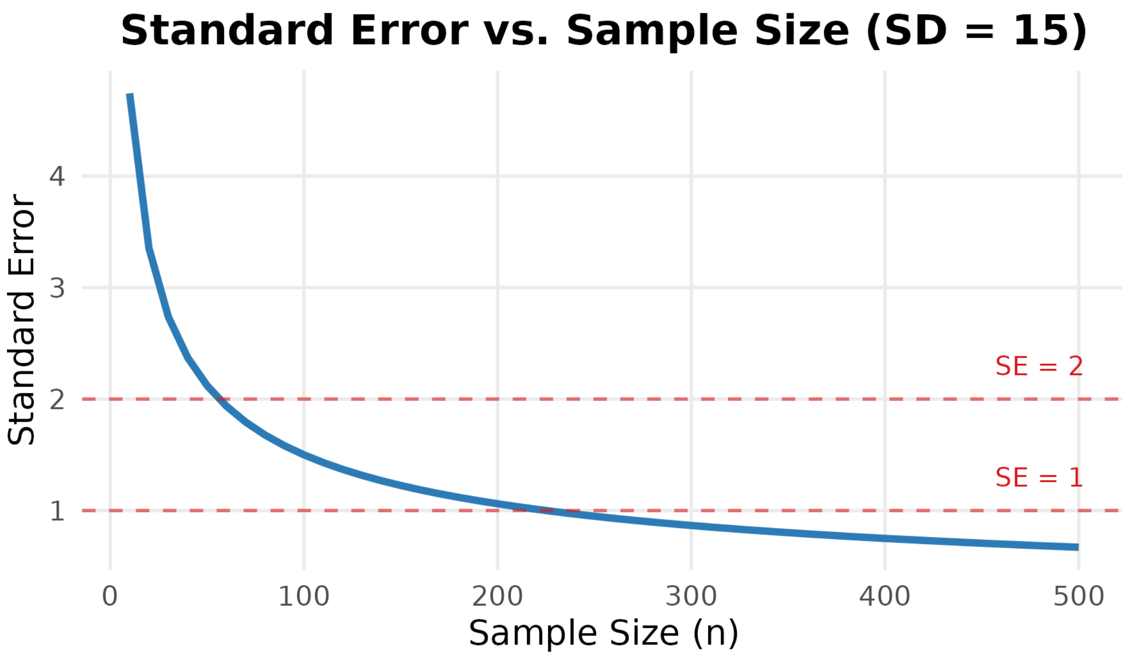

Figure 1: As sample size increases, the standard error decreases and estimates become more precise

Confidence Interval for a Mean

When the Population SD Is Known (Z-interval)

If the population standard deviation σ is known and either (a) the population is normal, or (b) the sample size is large ($n \geq 30$), we use the **Z-interval**:

$$\bar{x} \pm z_{1-\alpha/2} \cdot \frac{\sigma}{\sqrt{n}}$$

The **critical value** $z_{1-\alpha/2}$ is written using the **left cumulative probability** (the area to the *left* of the value, matching the standard normal table and SPSS `IDF.NORMAL`). It depends on the confidence level:

Table 1: Critical Values for Common Confidence Levels

Confidence Level	alpha	1 - alpha/2 (left area)	z(1-alpha/2)
90%	0.10	0.950	1.645
95%	0.05	0.975	1.960
99%	0.01	0.995	2.576

The Most Used Value

In health research, the 95% CI with $z_{0.975} = 1.96$ is by far the most commonly reported. You will see it in almost every clinical paper.

When the Population SD Is Unknown (t-interval)

In practice, σ is almost never known. When we use the sample SD s as an estimate, we must use the **t-distribution** instead of the Z-distribution:

$$\bar{x} \pm t_{1-\alpha/2,\, n-1} \cdot \frac{s}{\sqrt{n}}$$

Where $t_{1-\alpha/2,n-1}$ is the critical value from the **t-distribution** with $n-1$ **degrees of freedom (df)**. As with z, the subscript $1-\alpha/2$ is the **left cumulative probability** (e.g., 0.975 for a 95% CI).

The t-Distribution

The t-distribution is similar to the standard normal distribution but has **heavier tails** — reflecting the extra uncertainty from estimating σ with s. As n increases, the t-distribution approaches the Z-distribution.

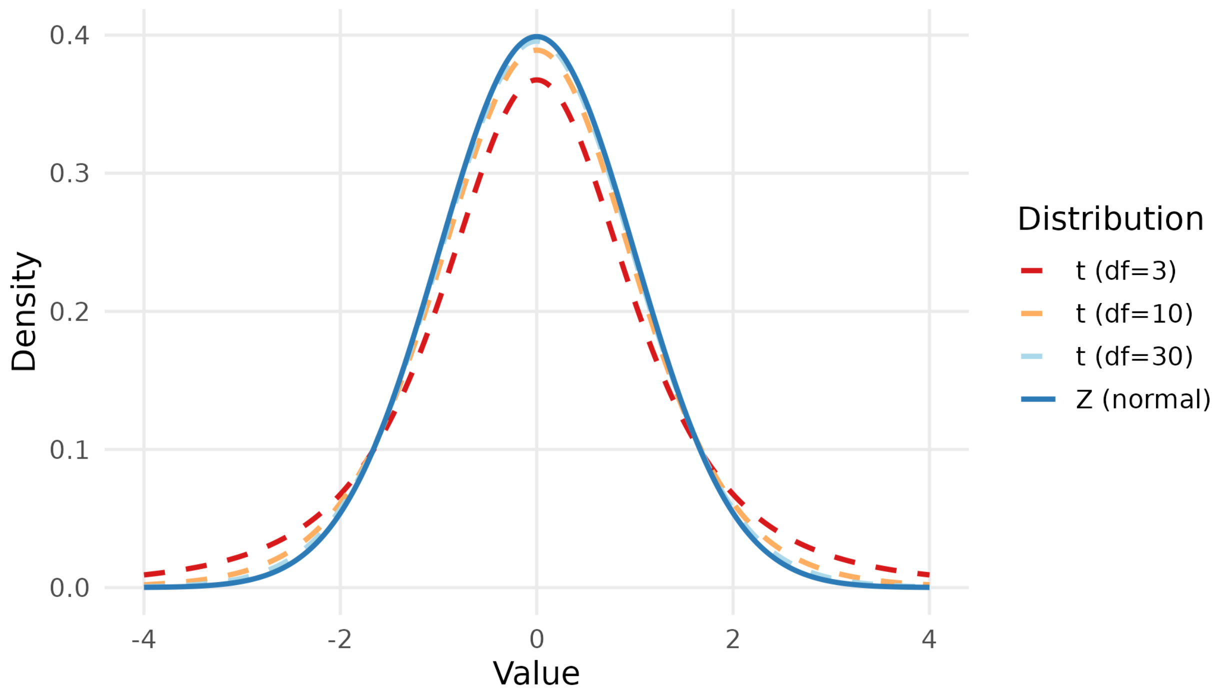

Figure 1: t-distributions with different degrees of freedom compared to Z

t Critical Values

Table 2: Selected t Critical Values (95% CI, two-tailed: left cumulative probability = 0.975)

Degrees of Freedom (df = n-1)	t(0.975, df)
5	2.571
10	2.228
15	2.131
20	2.086
25	2.060

30	2.042
40	2.021
60	2.000
120	1.980
Inf (Z)	**1.960**

As df increases, the t critical value converges to **1.960** (the Z value). For $n \geq 30$, the difference is small and Z is often used as an approximation.

Confidence Interval for a Proportion

When the outcome of interest is a **proportion** (e.g., prevalence of a disease, percentage of patients who respond to treatment), we use:

$$\hat{p} \pm z_{1-\alpha/2} \cdot \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$$

Where $\hat{p} = \frac{x}{n}$ is the sample proportion (x = number with the characteristic, n = total sample size).

> When Is This Formula Valid?
>
> This formula requires a **sufficiently large sample**. A common rule of thumb is:
>
> $$n\hat{p} \geq 5 \quad \text{and} \quad n(1-\hat{p}) \geq 5$$
>
> If this condition is not met, special methods are needed (not covered in this course).

Interpreting a Confidence Interval

Correct Interpretation

> *"We are 95% confident that the true population mean lies between [lower bound] and [upper bound]."*

What this means: If we repeated the sampling process many times, 95% of the resulting confidence intervals would contain the true parameter.
What this does NOT mean: There is NOT a 95% probability that the true parameter lies in this specific interval — the true parameter is fixed; it either is or is not in the interval.

The figure above shows 50 confidence intervals from 50 different samples of the same population. Most (95%) contain the true mean; a few do not — this is expected and not an error.

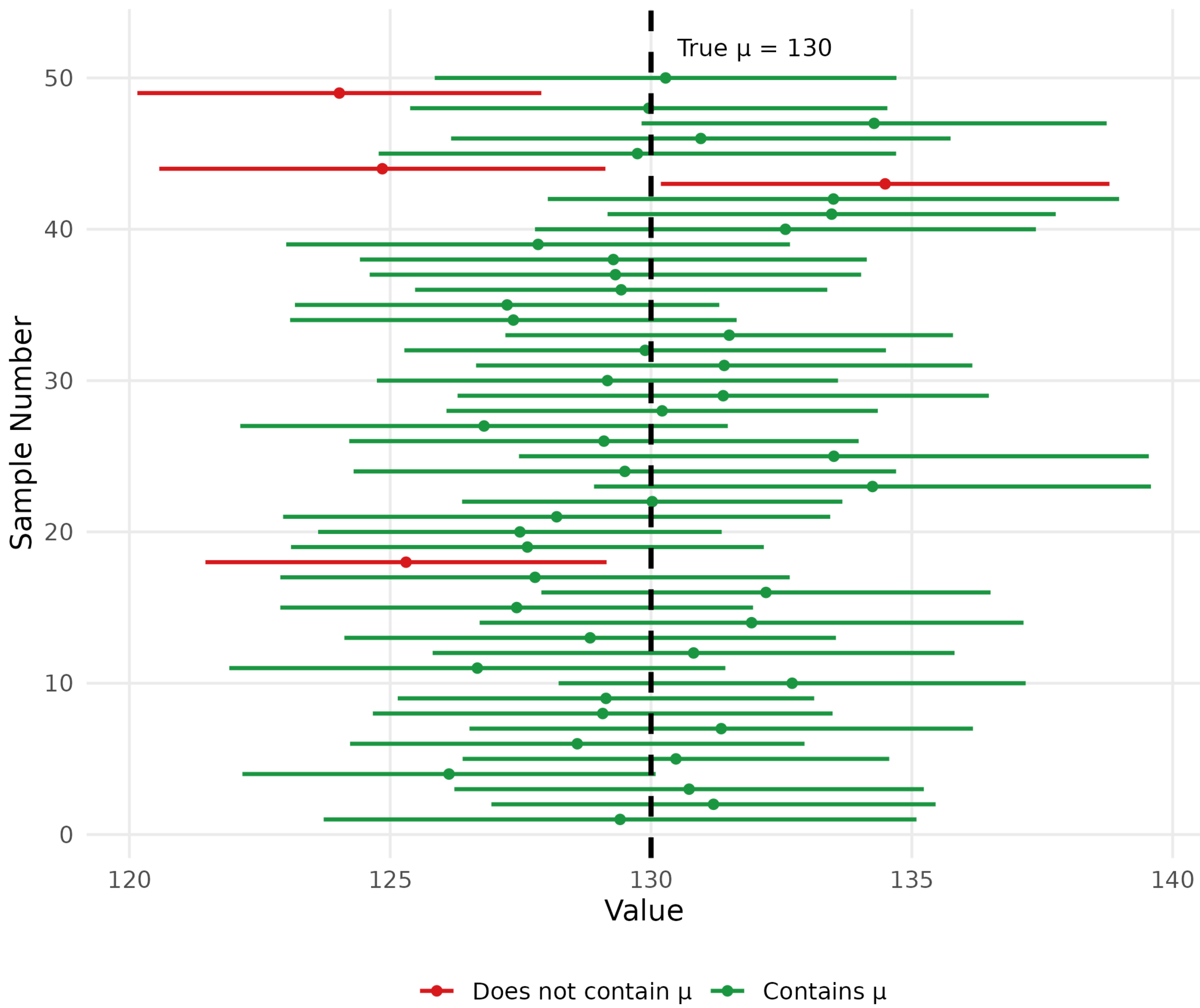

Figure 1: Simulation: 50 confidence intervals from repeated samples. Green intervals contain the true mean (= 130); red ones do not.

Key Takeaways

Chapter 4 Summary

1. A **point estimate** gives a single best-guess value for a population parameter; an **interval estimate** gives a range of plausible values.
2. A **confidence interval (CI)** = Point estimate ± Margin of error. The margin of error depends on the confidence level, the SE, and the critical value.
3. Use the **Z-interval** when σ is known or $n \geq 30$; use the **t-interval** when σ is unknown and n is small, with $df = n - 1$.
4. The **95% CI** is the standard in health research, using $z_{0.975} = 1.96$.
5. A wider CI means less precision; a larger sample size n always produces a narrower (more precise) CI.
6. The correct interpretation: *"We are 95% confident the true parameter lies within this interval"* — not a probability statement about a fixed parameter.

Examples

Example 1: 95% CI for a Mean (unknown)

A study measured the fasting blood glucose (mg/dL) of a random sample of **36 adults** and found:

$$\bar{x} = 105.4 \text{ mg/dL}, \qquad s = 18.6 \text{ mg/dL}$$

Construct a **95% confidence interval** for the population mean blood glucose.

Solution

Given: $n = 36$, $\bar{x} = 105.4$, $s = 18.6$, confidence level = 95%
Step 1 — Standard Error:

$$SE = \frac{s}{\sqrt{n}} = \frac{18.6}{\sqrt{36}} = \frac{18.6}{6} = 3.10$$

Step 2 — Critical value:
$df = n - 1 = 35$. From the t-table at left cumulative probability $1 - \alpha/2 = 0.975$: $t_{0.975,\,35} = 2.030$
(Since $n = 36 \geq 30$, we could also use $z = 1.96$ as an approximation.)
Step 3 — Confidence Interval:

$$105.4 \pm 2.030 \times 3.10 = 105.4 \pm 6.29$$

95% CI: (99.1, 111.7) mg/dL

Interpretation: We are 95% confident that the true mean fasting blood glucose of adults in this population is between **99.1 and 111.7 mg/dL**.

Example 2: 95% CI for a Proportion

In a survey of **200 adults**, **68 were found to have hypertension**.

Construct a **95% confidence interval** for the true prevalence of hypertension in this population.

Solution

Given: $n = 200$, $x = 68$
Step 1 — Sample proportion:

$$\hat{p} = \frac{68}{200} = 0.34 \quad (34\%)$$

Step 2 — Check validity:

$$n\hat{p} = 200 \times 0.34 = 68 \geq 5\checkmark$$

$$n(1 - \hat{p}) = 200 \times 0.66 = 132 \geq 5\checkmark$$

Step 3 — Standard Error:

$$SE = \sqrt{\frac{0.34 \times 0.66}{200}} = \sqrt{\frac{0.2244}{200}} = \sqrt{0.001122} = 0.0335$$

Step 4 — Confidence Interval (using $z_{0.975} = 1.96$):

$$0.34 \pm 1.96 \times 0.0335 = 0.34 \pm 0.066$$

$$\textbf{95\% CI:}\ (0.274,\ 0.406) \quad \text{or} \quad (27.4\%,\ 40.6\%)$$

Interpretation: We are 95% confident that the true prevalence of hypertension in this population is between **27.4% and 40.6%**.

Example 3: Effect of Sample Size on CI Width

Using the same data as Example 1 ($\bar{x} = 105.4$, $s = 18.6$), compare the 95% CI when $n = 36$, $n = 100$, and $n = 400$.

Solution

Table 1: Effect of Sample Size on CI Width (xbar = 105.4, s = 18.6, 95% CI)

n	SE	Margin of Error	Lower Bound	Upper Bound	CI Width
36	3.10	6.076	99.32	111.48	**12.15**
100	1.86	3.646	101.75	109.05	**7.29**
400	0.93	1.823	103.58	107.22	**3.65**

Conclusion: Quadrupling the sample size (from 100 to 400) **halves** the CI width — because SE depends on $\sqrt{n}$, not n directly. To halve the margin of error, you need to **quadruple** the sample size.

Exercises

Exercise 1

A random sample of **25 patients** with chronic kidney disease had their serum creatinine levels measured:

$$\bar{x} = 3.82 \text{ mg/dL}, \qquad s = 1.14 \text{ mg/dL}$$

(a) Calculate the standard error of the mean.
(b) Find the appropriate critical value for a 95% CI.
(c) Construct the **95% confidence interval** for the population mean serum creatinine level.
(d) Interpret the confidence interval in plain language.

Exercise 2

A hospital records that **45 out of 180 patients** discharged last month were readmitted within 30 days.
(a) Calculate the sample proportion of 30-day readmissions.
(b) Check whether the large-sample conditions are satisfied.
(c) Construct a **95% confidence interval** for the true 30-day readmission rate.
(d) The hospital's target readmission rate is 20%. Based on the confidence interval, is there evidence that the true rate differs from this target? Explain.

Exercise 3

A researcher wants to estimate the mean diastolic blood pressure (DBP) of adults in a city. From a pilot study, the SD is estimated to be **12 mmHg**.
The researcher wants a **95% CI with a margin of error no greater than 2 mmHg**.
Using the formula:

$$n = \left(\frac{z_{1-\alpha/2} \cdot \sigma}{E} \right)^2$$

where E is the desired margin of error, calculate the **minimum sample size** required.

End of Chapter 4

Coming Up Next

Chapter 5: Statistical Testing — Instead of estimating a parameter, we will test specific claims about populations using **hypothesis testing** — one of the most powerful tools in health research.

Statistical Testing

Introduction

Why Does This Matter?

In health research, we constantly ask questions that require a decision:

- Does this new drug lower blood pressure *more* than the standard treatment?
- Is there a relationship between smoking and lung disease?
- Do recovery times differ between two surgical techniques?

We cannot answer these questions simply by looking at sample means or proportions — random chance alone can create apparent differences between groups. **Statistical testing** gives us a formal, objective framework for deciding whether an observed difference is likely to be real or simply due to chance.

This is the most widely used tool in health research, and understanding it is essential for reading and interpreting scientific literature.

Concept of Hypothesis Testing

The Logic of Hypothesis Testing

Statistical testing works by assuming the opposite of what we want to prove, then asking: *"If nothing were really happening, how surprising would our data be?"*

If the data would be very surprising under the assumption that nothing is happening, we conclude that something probably *is* happening.

The Two Hypotheses

Every hypothesis test involves two competing statements:

Null Hypothesis (H_0)

The **null hypothesis** states that there is **no effect, no difference, or no relationship** in the population. It is the assumption we start with and attempt to disprove.

> *Examples:* - The mean blood pressure is 120 mmHg. ($H_0 : \mu = 120$) - There is no difference in recovery time between the two groups. ($H_0 : \mu_1 = \mu_2$) - There is no association between smoking status and disease. (H_0 : no association)

Alternative Hypothesis (H_1 or H_a)

The **alternative hypothesis** states that there **is** an effect, difference, or relationship. This is usually what the researcher believes or wants to demonstrate.

> *Examples:* - The mean blood pressure is not 120 mmHg. ($H_1 : \mu \neq 120$) - Recovery time differs between the two groups. ($H_1 : \mu_1 \neq \mu_2$) - There is an association between smoking and disease. (H_1 : association exists)

One-Tailed vs. Two-Tailed Tests

The alternative hypothesis can be **directional** or **non-directional**:

Table 1: One-tailed vs. Two-tailed Tests

Test Type	H1	Use When
Two-tailed	mu != mu0 (different from)	You want to detect a difference in either dir
One-tailed (right)	mu > mu0 (greater than)	You specifically expect the value to be highe
One-tailed (left)	mu < mu0 (less than)	You specifically expect the value to be lower

Practical Advice

In most health research, use a **two-tailed test** unless you have a strong, pre-specified reason to test in only one direction. Two-tailed tests are more conservative and more widely accepted.

The p-value

The **p-value** is the probability of obtaining results *at least as extreme as those observed*, assuming the null hypothesis is true.

$$p\text{-value} = P(\text{observed result or more extreme} \mid H_0 \text{ is true})$$

- A **small p-value** means the observed data would be very unlikely if H_0 were true evidence against H_0
- A **large p-value** means the observed data are consistent with H_0 insufficient evidence to reject H_0

The Significance Level ( )

The **significance level** α is the threshold we set in advance for deciding whether the p-value is "small enough" to reject H_0.

The most common choice in health research is $\alpha = 0.05$.

Decision Rule	
If p-value $\leq \alpha$	**Reject** H_0 — result is statistically significant
If p-value $> \alpha$	**Fail to reject** H_0 — insufficient evidence

> Common Misinterpretations
>
> **"Failing to reject H_0" is NOT the same as proving H_0 is true.** It simply means the data do not provide strong enough evidence against it.
> **"Statistically significant" does NOT necessarily mean clinically important.** With a very large sample, even a trivially small difference can become statistically significant. Always consider the practical meaning of results.

Type I and Type II Errors

When making a decision about H_0, two types of errors are possible:

Table 2: Type I and Type II Errors in Hypothesis Testing

Our Decision	H0 is TRUE	H0 is FALSE
Reject H0	Type I Error (alpha) - False Positive	Correct Decision (Power = 1 - beta
Fail to reject H0	Correct Decision	Type II Error (beta) - False Negativ

- **Type I error ( ):** Concluding there is an effect when there is none (*false positive*). Controlled by choosing (usually 0.05).
- **Type II error ( ):** Missing a real effect (*false negative*). The probability of correctly detecting a real effect is called **power** $= 1 - \beta$.

General Steps in Hypothesis Testing

The 5-Step Framework

Step 1 — State the hypotheses: Write H_0 and H_1 clearly.
Step 2 — Choose the significance level: Usually $\alpha = 0.05$.
Step 3 — Select and compute the test statistic: A number calculated from the sample that measures how far the data are from what H_0 predicts.
Step 4 — Find the p-value: The probability of getting a test statistic this extreme if H_0 is true.
Step 5 — Make a decision and interpret: Compare p-value to . State your conclusion in the context of the problem.

Statistical Testing on Categorical Data

Chi-Square Test for Two Independent Groups

When to Use

The **chi-square test of independence** is used when:

- The outcome variable is **categorical (nominal)**
- We are comparing **two or more independent groups**
- We want to test whether there is an **association** between two categorical variables

The Contingency Table

Data are arranged in a **contingency table** (also called a cross-tabulation):

	Outcome: Yes	**Outcome: No**	**Row Total**
Group 1	a	b	$a+b$
Group 2	c	d	$c+d$
Column Total	$a+c$	$b+d$	n

The Test Statistic

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

Where: - O = observed frequency (actual count in each cell) - E = expected frequency = $\frac{\text{Row total} \times \text{Column total}}{n}$

The sum is taken over **all cells** in the table.

The test statistic follows a **chi-square distribution** with $df = (\text{rows}-1) \times (\text{columns}-1)$.

For a 2×2 table: $df = (2-1)(2-1) = 1$.

Condition for Valid Chi-Square Test

All **expected frequencies** must be ≥ 5. If this condition is violated, use **Fisher's exact test** instead (available in SPSS).

Chi-Square Critical Values

Table 1: Chi-Square Critical Values at alpha = 0.05

Degrees of Freedom (df)	chi2 critical value (alpha = 0.05)
1	3.841
2	5.991
3	7.815
4	9.488
5	11.070
6	12.592
7	14.067
8	15.507
9	16.919
10	18.307

Statistical Testing on Rank Data

Mann-Whitney U Test for Two Independent Groups

When to Use

The **Mann-Whitney U test** (also called the Wilcoxon rank-sum test) is used when:

- The outcome variable is **ordinal**, or
- The outcome is **continuous but not normally distributed** (especially with small samples or obvious skewness)
- We are comparing **two independent groups**

It is a **non-parametric** test — it does not assume the data follow any particular distribution.

How It Works

Instead of comparing means, the Mann-Whitney test compares the **ranks** of observations between the two groups.

Steps:

1. Combine all observations from both groups and **rank them** from smallest (rank 1) to largest.
2. If there are **tied values**, assign the average rank to each tied observation.
3. Calculate the **sum of ranks** for each group: W_1 and W_2.
4. The test statistic U is computed from the rank sums.
5. Compare to the critical value or obtain the p-value from software (SPSS).

> Intuition Behind the Test
>
> If the two groups come from the same distribution, their ranks should be **randomly mixed** between the groups — neither group should consistently have higher or lower ranks. A significant result means one group consistently has higher ranks (i.e., higher values) than the other.

The U Statistic

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - W_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - W_2$$

$$U = \min(U_1, U_2)$$

Where n_1, n_2 are the sample sizes and W_1, W_2 are the rank sums for groups 1 and 2 respectively.

For large samples, U is approximately normally distributed and a Z-score can be computed. In practice, **SPSS computes the p-value directly**.

Wilcoxon Signed-Ranks Test for Two Related Groups

When to Use

The **Wilcoxon signed-ranks test** (also called the Wilcoxon matched-pairs test) is the non-parametric alternative to the **paired samples t-test**. It is used when:

- The outcome variable is **ordinal**, or
- The **differences** between pairs are **not normally distributed** (especially with small samples)
- The two sets of measurements are **related (paired)** — for example, the same patients measured before and after treatment, or two measurements taken on the same blood sample

> Mann-Whitney or Wilcoxon?
>
> Both tests work with ranks, so they are easy to confuse. The difference is the **study design**, exactly as with the two t-tests:
>
> - **Two independent groups** (different people in each group) → **Mann-Whitney U test**
> - **Two related groups** (the same people measured twice) → **Wilcoxon signed-ranks test**

How It Works

The Wilcoxon test looks at the **differences within each pair**, and then ranks the **sizes** of those differences.

Steps:

1. For each pair, calculate the difference $d_i = x_{i,1} - x_{i,2}$.
2. **Discard** any pair whose difference is zero.
3. Rank the **absolute** differences $|d_i|$ from smallest to largest, assigning average ranks to ties.
4. Attach to each rank the **sign** of its original difference.
5. Add the ranks belonging to the positive differences (W^+) and the ranks belonging to the negative differences (W^-).

The Test Statistic

$$W = \min(W^+,\ W^-)$$

If the two conditions really are the same, the positive and negative ranks should roughly balance, so W^+ and W^- should be similar. A **small** value of W means the differences fall mostly in one direction, which is evidence against H_0.

For larger samples the test statistic is converted to a Z-score, and this is what **SPSS reports** together with the p-value:

$$H_0 : \text{the two related measurements are equal}$$
$$H_1 : \text{the two related measurements are different}$$

Reading the SPSS Output

Choose **Analyze** → **Nonparametric Tests** → **2 Related Samples**, and tick **Wilcoxon**. The *Test Statistics* table reports Z and the 2-tailed significance. As always, decide by comparing the p-value with α — the sign of Z only indicates the direction of the ranks, not the importance of the result.

Worked Example

Blood clotting time was measured for **8 patients** using two solutions. Blood was taken from each patient and divided into two parts, so the two measurements are **paired**.

Table 1: Blood clotting time using two solutions (n = 8 patients)

Patient	Shrimp-shell extract	Standard solution	Difference d
1	3	7	-4
2	5	6	-1
3	5	3	2
4	4	8	-4
5	3	5	-2
6	7	9	-2
7	8	7	1
8	7	9	-2

Because the sample is small ($n = 8$) and the outcome is a clotting time that need not be normally distributed, the **Wilcoxon signed-ranks test** is used rather than the paired t-test. SPSS reports:

$$Z = -1.706, \qquad p\text{-value} = 0.088$$

Since $p = 0.088 > \alpha = 0.05$, we **do not reject** H_0. There is not enough evidence to say that the clotting times produced by the two solutions differ, at the 0.05 level of significance.

Statistical Testing on Quantitative Data

Independent Samples t-Test

When to Use

The **independent samples t-test** is used when:

- The outcome variable is **continuous (numerical)**
- We are comparing the **means of two independent groups**
- The data are approximately **normally distributed** in each group (or sample sizes are large enough, $n \geq 30$ per group)

Hypotheses

$$H_0 : \mu_1 = \mu_2 \quad \text{(no difference in population means)}$$
$$H_1 : \mu_1 \neq \mu_2 \quad \text{(the population means differ)}$$

The Test Statistic

$$t = \frac{\bar{x}_1 - \bar{x}_2}{SE_{\bar{x}_1 - \bar{x}_2}}$$

The standard error of the difference depends on whether the two groups have **equal or unequal variances** — SPSS performs **Levene's test** automatically to help you decide which version to use.

Equal variances assumed (pooled):

$$SE = s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}, \qquad s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

Equal variances not assumed (Welch's):

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

The test statistic follows a **t-distribution** with $df = n_1 + n_2 - 2$ (equal variances) or an approximated df (Welch's).

Paired Samples t-Test

When to Use

The **paired samples t-test** is used when:

- The outcome variable is **continuous (numerical)**
- Observations come in **natural pairs** — the same subject measured twice (e.g., before and after treatment), or matched pairs of subjects
- We want to test whether the **mean difference** within pairs is zero

Hypotheses

$$H_0 : \mu_d = 0 \quad \text{(mean difference is zero)}$$
$$H_1 : \mu_d \neq 0 \quad \text{(mean difference is not zero)}$$

Where μ_d is the population mean of the within-pair differences.

The Test Statistic

For each pair i, compute the difference: $d_i = x_{1i} - x_{2i}$

Then:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}}$$

Where: - $\bar{d}$ = mean of the differences - s_d = standard deviation of the differences - n = number of pairs

The test statistic follows a **t-distribution** with $df = n - 1$.

Choosing the Right Test

Table 1: Guide to Choosing the Correct Statistical Test (Tw

Outcome Data Type	Study Design	Statistical Test	SPSS P
Categorical (Nominal)	2 independent groups	**Chi-square test**	Analyz
Ordinal or non-normal continuous	2 independent groups	**Mann-Whitney U test**	Analyz
Ordinal or non-normal continuous	2 related (paired) groups	**Wilcoxon signed-ranks test**	Analyz
Continuous (normal)	2 independent groups	**Independent samples t-test**	Analyz
Continuous (normal)	2 related (paired) groups	**Paired samples t-test**	Analyz

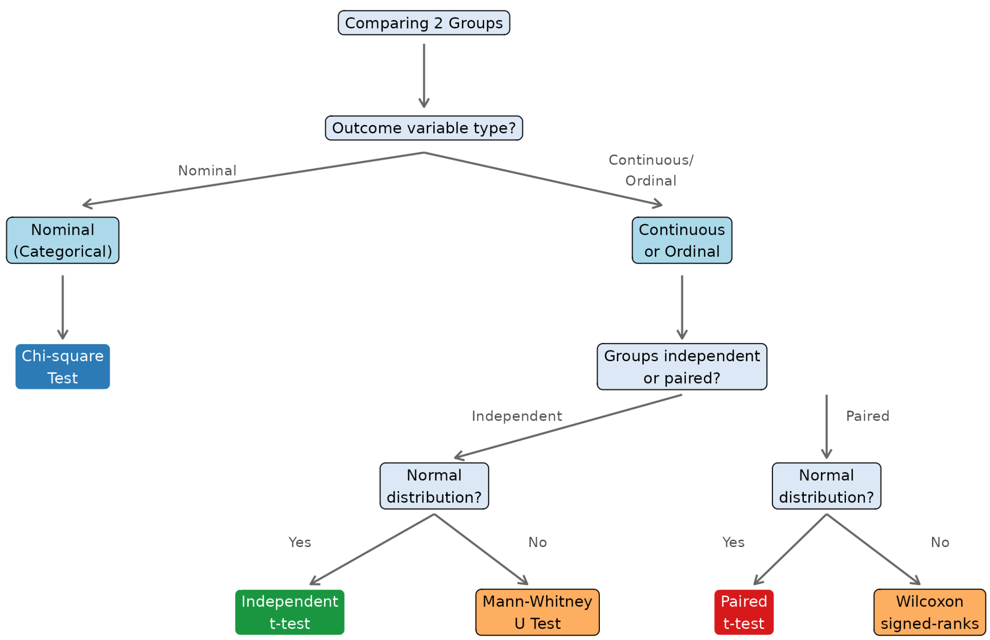

Figure 1: Decision flowchart for selecting the correct test (two-group comparisons)

Key Takeaways

Chapter 5 Summary

1. Hypothesis testing uses sample data to decide between two competing claims: the **null hypothesis** (H_0) and the **alternative hypothesis** (H_1).
2. The **p-value** is the probability of the observed data (or more extreme) if H_0 is true. A p-value $\leq \alpha$ (usually 0.05) leads us to **reject** H_0.
3. **Type I error ( ):** rejecting a true H_0 (false positive). **Type II error ( ):** failing to reject a false H_0 (false negative).
4. **Chi-square test:** for association between two categorical (nominal) variables in independent groups.
5. **Mann-Whitney U test:** for comparing two independent groups when the outcome is ordinal or the normality assumption is not met.
6. **Wilcoxon signed-ranks test:** for comparing two **related** measurements when the outcome is ordinal or the differences are not normally distributed. It is the non-parametric alternative to the paired t-test.
7. **Independent samples t-test:** for comparing means of two independent groups with continuous, approximately normal data.
8. **Paired samples t-test:** for comparing means of two related measurements (before/after, or matched pairs) with continuous data.
9. The two rank tests differ by **design**: Mann-Whitney for *independent* groups, Wilcoxon signed-ranks for *related* pairs.
10. Always choose the test based on **data type** and **study design** — using the wrong test leads to invalid conclusions.

Examples

Example 1: Chi-Square Test

A study investigates whether **smoking status** is associated with **hypertension** in a sample of 200 adults.

	Hypertension	No Hypertension	Total
Smoker	45	55	100
Non-smoker	25	75	100
Total	**70**	**130**	**200**

Test at $\alpha = 0.05$ whether smoking is associated with hypertension.

Solution

Step 1 — Hypotheses:
H_0: Smoking status and hypertension are independent (no association).
H_1: Smoking status and hypertension are associated.
Step 2 — $\alpha = 0.05$
Step 3 — Expected frequencies:

$$E = \frac{\text{Row total} \times \text{Column total}}{n}$$

Table 1: Observed (O) and Expected (E) Frequencies

Group	O (Hyp.)	E (Hyp.)	O (No Hyp.)	E (No Hyp.)
Smoker	45	35	55	65
Non-smoker	25	35	75	65

All expected frequencies ≥ 5
Step 4 — Test statistic:

$$\chi^2 = \frac{(45-35)^2}{35} + \frac{(55-65)^2}{65} + \frac{(25-35)^2}{35} + \frac{(75-65)^2}{65}$$

$$= \frac{100}{35} + \frac{100}{65} + \frac{100}{35} + \frac{100}{65} = 2.857 + 1.538 + 2.857 + 1.538 = \mathbf{8.791}$$

$df = (2-1)(2-1) = 1$
Step 5 — Decision:
From the chi-square table: $\chi^2_{\text{critical}} = 3.841$ at $df = 1$, $\alpha = 0.05$.
Since $8.791 > 3.841$, we **reject** H_0.
The p-value $= 0.003 < 0.05$.
Conclusion: There is a statistically significant association between smoking status and hypertension ($\chi^2 = 8.79$, $df = 1$, $p = 0.003$). Smokers have a higher prevalence of hypertension than non-smokers.

Example 2: Independent Samples t-Test

A clinical trial compares the mean reduction in systolic blood pressure (mmHg) between patients receiving a **new drug** vs. a **placebo** after 8 weeks.

Group	n	Mean reduction	SD
New drug	40	12.4	6.8
Placebo	40	7.1	7.2

Test at $\alpha = 0.05$ whether the new drug produces a greater mean reduction in SBP than placebo.

Solution

Step 1 — Hypotheses:
$H_0 : \mu_1 = \mu_2$ (no difference in mean reduction)
$H_1 : \mu_1 \neq \mu_2$ (two-tailed; difference exists)
Step 2 — $\alpha = 0.05$
Step 3 — Pooled standard error (assuming equal variances):

$$s_p = \sqrt{\frac{(40-1)(6.8)^2 + (40-1)(7.2)^2}{40+40-2}} = \sqrt{\frac{39(46.24) + 39(51.84)}{78}} = \sqrt{\frac{1803.36 + 2021.76}{78}} = \sqrt{49.041} =$$

$$SE = 7.003 \times \sqrt{\frac{1}{40} + \frac{1}{40}} = 7.003 \times \sqrt{0.05} = 7.003 \times 0.2236 = 1.566$$

Step 4 — Test statistic:

$$t = \frac{12.4 - 7.1}{1.566} = \frac{5.3}{1.566} = \mathbf{3.384}$$

$df = 40 + 40 - 2 = 78$

Step 5 — Decision:

$t_{\text{critical}} = 1.991$ at $df = 78$, $\alpha = 0.05$ (two-tailed).

Since $3.384 > 1.991$, we **reject** H_0.

The p-value $= 0.001 < 0.05$.

Conclusion: The new drug produces a significantly greater reduction in systolic blood pressure compared to placebo ($t = 3.38$, $df = 78$, $p = 0.001$). The mean difference is 5.3 mmHg (95% CI: 2.2 to 8.4 mmHg).

Example 3: Paired Samples t-Test

A study measures the **diastolic blood pressure (DBP)** of 10 patients **before** and **after** 4 weeks of a lifestyle intervention.

Table 2: DBP Before and After Lifestyle Intervention (n = 10 patients)

Patient	DBP Before (mmHg)	DBP After (mmHg)	Difference d = Before - Aft
Patient 1	92	85	**7**
Patient 2	88	82	**6**
Patient 3	95	89	**6**
Patient 4	100	91	**9**
Patient 5	85	80	**5**
Patient 6	97	88	**9**
Patient 7	90	84	**6**
Patient 8	94	87	**7**
Patient 9	88	83	**5**
Patient 10	96	88	**8**

Test at $\alpha = 0.05$ whether the intervention significantly reduces DBP.

Solution

Step 1 — Hypotheses:
$H_0 : \mu_d = 0$ (no mean change in DBP)
$H_1 : \mu_d \neq 0$ (mean DBP has changed)
Step 2 — $\alpha = 0.05$
Step 3 — Compute $\bar{d}$ and s_d:

$$\bar{d} = \frac{68}{10} = 6.8 \text{ mmHg}$$

$$s_d = \sqrt{\frac{\sum(d_i - \bar{d})^2}{n-1}} = 1.476$$

Step 4 — Test statistic:

$$t = \frac{\bar{d}}{s_d/\sqrt{n}} = \frac{6.8}{1.476/\sqrt{10}} = \frac{6.8}{0.467} = 14.571$$

$df = n - 1 = 9$
Step 5 — Decision:
$t_{\text{critical}} = 2.262$ at $df = 9$, $\alpha = 0.05$ (two-tailed).
Since $14.571 > 2.262$, we **reject** H_0.
Conclusion: The lifestyle intervention produced a statistically significant reduction in diastolic blood pressure ($t = 14.57$, $df = 9$, $p < 0.05$). The mean reduction was **6.8 mmHg**.

Exercises

Exercise 1 — Chi-Square Test

A researcher investigates whether **physical activity level** is associated with **diabetes status** in 300 adults:

	Diabetes	No Diabetes	Total
Active	18	132	150
Inactive	42	108	150
Total	**60**	**240**	**300**

(a) State H_0 and H_1.
(b) Calculate all expected frequencies and verify the chi-square condition is met.
(c) Calculate the chi-square test statistic.
(d) At $\alpha = 0.05$, what is your conclusion?
(e) In plain language, what does your result mean?

Exercise 2 — Independent Samples t-Test

A study compares the **mean hemoglobin levels (g/dL)** between male and female patients:

Group	n	$\bar{x}$	s
Male	35	14.8	1.6
Female	35	12.9	1.4

(a) State H_0 and H_1 (two-tailed).
(b) Calculate the pooled standard error.
(c) Calculate the t test statistic.
(d) At $\alpha = 0.05$ with appropriate df, state your conclusion.

Exercise 3 — Paired Samples t-Test

A pain management study measures the **pain score (0–10)** of 8 patients **before** and **one hour after** receiving an analgesic:

Patient	1	2	3	4	5	6	7	8
Before	7	8	6	9	7	8	5	9
After	4	5	4	6	3	5	3	6

(a) Calculate the difference d_i for each patient.
(b) Calculate $\bar{d}$ and s_d.
(c) Calculate the paired t statistic.
(d) At $\alpha = 0.05$, is the analgesic effective in reducing pain? State your conclusion clearly.

Exercise 4 — Choosing the Right Test

For each scenario below, state which statistical test is most appropriate and briefly explain why:
(a) Comparing the proportion of post-operative complications between patients who received standard care vs. enhanced recovery protocol.
(b) Comparing patient satisfaction scores (rated: Poor / Fair / Good / Excellent) between two hospital wards.
(c) Measuring the effect of a new diet on body weight by recording each participant's weight before and after a 12-week program.
(d) Comparing the mean length of hospital stay (days) between two age groups (under 60 vs. 60 and over), where length-of-stay data is strongly right-skewed.
(e) Comparing pain scores (rated 0–10) recorded for the **same** patients before and 24 hours after starting a new analgesic, where the differences are clearly not normally distributed.

End of Chapter 5

Coming Up Next

Chapter 6: Introduction to Statistical Modeling — We take our first steps beyond comparing two groups, exploring how one variable can be used to explain or predict another through **linear regression**.

Introduction to Statistical Modeling

Introduction

Why Does This Matter?

So far we have described data, estimated population parameters, and tested whether two groups differ. But many real-world health questions go beyond simple comparisons:

- Does body weight *predict* blood pressure?
- As age *increases*, how does cholesterol *change*?
- Can we use a patient's BMI to *estimate* their fasting glucose level?

These questions require us to model the **relationship between two variables**. **Statistical modeling** is the process of building a mathematical equation that describes how one variable is related to another — and using that equation to understand, explain, and predict outcomes.

In this chapter we focus on the simplest and most widely used model: **simple linear regression**.

What Is a Statistical Model?

A **statistical model** is a mathematical equation that describes the relationship between:

- An **outcome variable** (also called the *dependent variable*, *response variable*, or Y) — the variable we want to explain or predict.
- One or more **predictor variables** (also called *independent variables*, *explanatory variables*, or X) — the variables we use to explain or predict Y.

In this chapter we study **simple linear regression** — one outcome Y and one predictor X, with a straight-line relationship.

Correlation: Measuring the Relationship

Before building a model, we first ask: *Is there a relationship between X and Y, and how strong is it?*

The Scatter Plot

A **scatter plot** displays pairs of values (x_i, y_i) as points on a graph. The pattern of points reveals the **direction** and **strength** of the relationship.

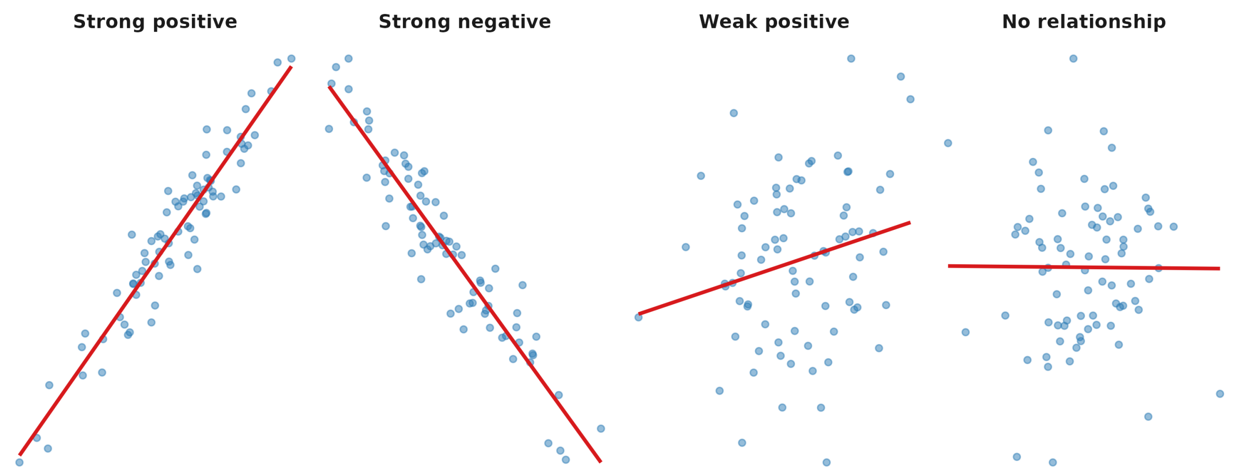

Figure 1: Four possible relationships between two variables

Pearson's Correlation Coefficient (r)

The **Pearson correlation coefficient** r quantifies the **strength** and **direction** of the *linear* relationship between two continuous variables.

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \cdot \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Properties of r

- r always lies between **-1 and $+1$**
- $r = +1$: perfect positive linear relationship
- $r = -1$: perfect negative linear relationship
- $r = 0$: no linear relationship

Interpreting the Magnitude of r

Table 1: General guidelines for interpreting the correlation coefficient

|r| value	Strength	Note
0.00 - 0.19	Negligible	-
0.20 - 0.39	Weak	Positive or negative depending on sign
0.40 - 0.59	Moderate	Positive or negative depending on sign
0.60 - 0.79	Strong	Positive or negative depending on sign
0.80 - 1.00	Very strong	Positive or negative depending on sign

Correlation Is Not Causation

A high correlation between X and Y does **not** mean that X *causes* Y. Both variables may be influenced by a third variable, or the relationship may be coincidental. Always interpret correlations in the context of biological or clinical plausibility.

Simple Linear Regression

The Regression Equation

Simple linear regression models the relationship between X and Y as a straight line:

$$\hat{Y} = b_0 + b_1 X$$

Where:

Symbol	Name	Meaning
$\hat{Y}$	Predicted value	The estimated value of Y for a given X
b_0	Intercept	The predicted value of Y when $X = 0$
b_1	Slope	The change in $\hat{Y}$ for each one-unit increase in X

The full model including random error is:

$$Y = b_0 + b_1 X + \varepsilon$$

Where ε represents the **residual** — the difference between the observed value and the predicted value:

$$\varepsilon_i = Y_i - \hat{Y}_i$$

Estimating the Coefficients

The values of b_0 and b_1 are chosen to minimize the **sum of squared residuals** — this method is called **Ordinary Least Squares (OLS)**:

$$\text{Minimize: } \sum_{i=1}^{n}(Y_i - \hat{Y}_i)^2 = \sum_{i=1}^{n}\varepsilon_i^2$$

The OLS formulas are:

$$b_1 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = r \cdot \frac{s_y}{s_x}$$

$$b_0 = \bar{y} - b_1\bar{x}$$

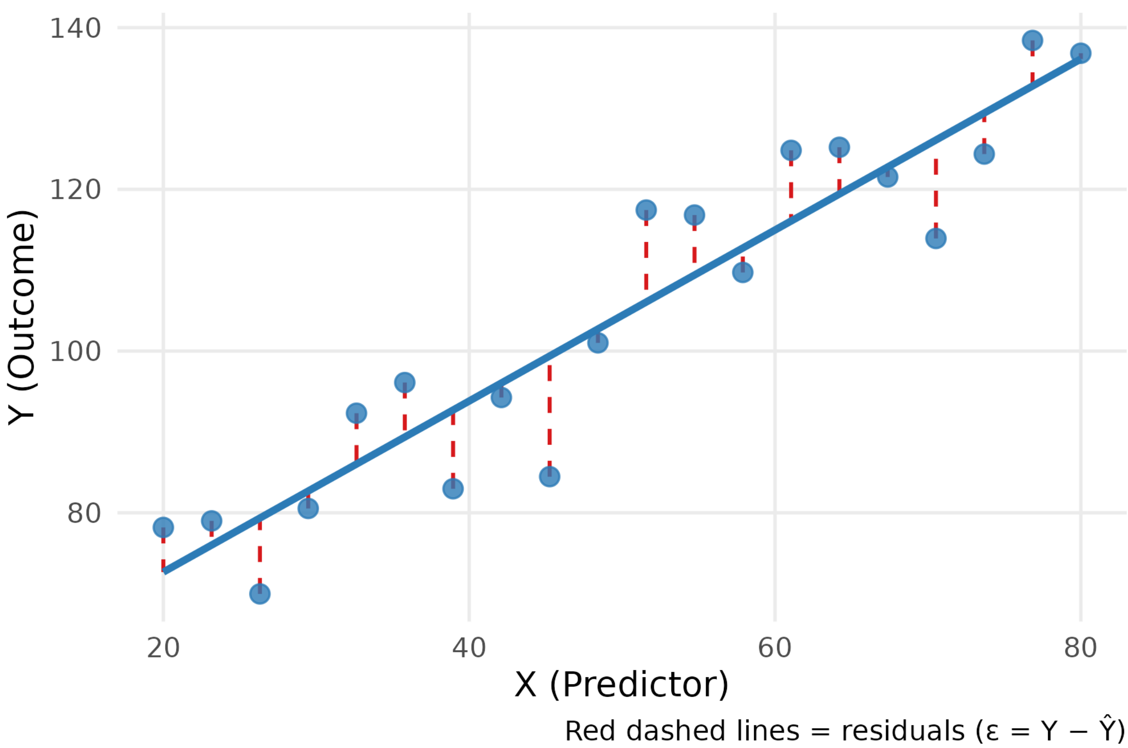

Figure 1: Least squares regression line minimizes the sum of squared residuals (vertical red lines)

The Coefficient of Determination (R^2)

R^2 (R-squared) measures how well the regression line fits the data — specifically, the **proportion of the total variation in Y that is explained by X**:

$$R^2 = r^2 \quad \text{(in simple linear regression)}$$

$$R^2 = \frac{\text{Variation explained by the model}}{\text{Total variation in } Y}$$

- $R^2 = 1$: the model explains 100% of the variation in Y (perfect fit)
- $R^2 = 0$: the model explains none of the variation in Y
- $R^2 = 0.65$ means: *"65% of the variation in Y is explained by X"*

Testing the Significance of the Regression

Before using the regression equation for interpretation or prediction, we should test whether the slope b_1 is significantly different from zero. If $b_1 = 0$, then X provides no information about Y.

Hypotheses:

$$H_0 : \beta_1 = 0 \quad \text{(X is not a useful predictor of Y)}$$
$$H_1 : \beta_1 \neq 0 \quad \text{(X is a useful predictor of Y)}$$

The test statistic is:

$$t = \frac{b_1}{SE_{b_1}}$$

which follows a t-distribution with $df = n - 2$.

SPSS reports this p-value automatically in the **Coefficients table** of the regression output.

Assumptions of Linear Regression

The regression model is valid only when the following assumptions are met:

Table 1: Assumptions of Simple Linear Regres

Assumption	Meaning
Linearity	The relationship between X and Y is linear (a straight line is approp
Independence	Observations are independent of each other
Homoscedasticity	The spread of residuals is roughly constant across all values of X
Normality of residuals	Residuals are approximately normally distributed

---

Reading SPSS Regression Output

SPSS produces three key tables for simple linear regression. Here is how to read them:

Table 2: SPSS Output: Model Summary

R	R2	Adjusted R2	Std. Error of the Estimate
0.821	0.674	0.666	8.432

$R = 0.821$ (strong positive correlation); **$R^2 = 0.674$** (67.4% of variation in Y explained by X).

Table 3: SPSS Output: ANOVA Table

	Sum of Squares	df	Mean Square	F	Sig.
Regression	**5621.4**	**1**	**5621.4**	**78.62**	**< 0.001**
Residual	2718.6	38	71.5		
Total	8340.0	39			

$F = 78.62$, $p < 0.001$ — the overall regression model is statistically significant.

Table 4: SPSS Output: Coefficients Table

	B	Std. Error	Beta (standardized)	t	Sig.
Constant	32.15	6.24	-	5.15	< 0.001

X (predictor)	**0.87**	**0.10**	**0.821**	**8.87**	**< 0.001**

Regression equation: $\hat{Y} = 32.15 + 0.87X$

The slope $\mathbf{B = 0.87}$ is statistically significant ($p < 0.001$): for each one-unit increase in X, Y increases by **0.87 units** on average.

Key Takeaways

Chapter 6 Summary

1. **Statistical modeling** describes the mathematical relationship between an outcome variable Y and one or more predictor variables X.
2. **Pearson's** r measures the strength and direction of a linear relationship between two continuous variables ($-1 \le r \le +1$).
3. **Simple linear regression** fits the equation $\hat{Y} = b_0 + b_1 X$ using the least squares method, which minimizes the sum of squared residuals.
4. The **slope** b_1 tells us how much Y changes for each one-unit increase in X; the **intercept** b_0 is the predicted value of Y when $X = 0$.
5. R^2 is the proportion of variation in Y explained by the model. A higher R^2 means a better fit.
6. Always test whether b_1 is significantly different from zero ($H_0 : \beta_1 = 0$) before using the model for interpretation.
7. The model is only valid when the four assumptions — **linearity, independence, homoscedasticity, and normality of residuals** — are reasonably satisfied.

Examples

Example 1: Calculating Correlation and Regression

A researcher records the **age (years)** and **systolic blood pressure (mmHg)** of 8 adults:

Table 1: Age and Systolic Blood Pressure of 8 Adults

Subject	Age (years)	SBP (mmHg)
S1	25	115
S2	32	120
S3	40	128
S4	48	135
S5	55	142
S6	62	148
S7	70	158
S8	78	165

(a) Calculate Pearson's r.

(b) Find the regression equation $\hat{Y} = b_0 + b_1 X$.

(c) Predict the SBP of a 50-year-old adult.

(d) Interpret R^2.

Solution

Table 2: Calculation Table for Correlation and Regression

Subject	x	y	x-xbar	y-ybar	(x-xbar)2	(y-ybar)2	(x-xbar)(y-ybar)
S1	25	115	-26.25	-23.88	689.06	570.02	626.72
S2	32	120	-19.25	-18.88	370.56	356.27	363.34
S3	40	128	-11.25	-10.88	126.56	118.27	122.34
S4	48	135	-3.25	-3.88	10.56	15.02	12.59
S5	55	142	3.75	3.12	14.06	9.77	11.72

S6	62	148	10.75	9.12	115.56	83.27	98.09
S7	70	158	18.75	19.12	351.56	365.77	358.59
S8	78	165	26.75	26.12	715.56	682.52	698.84
****Total****	**NA**	**NA**	**NA**	**NA**	**2393.50**	**2200.88**	**2292.25**

(a) Pearson's r:

$$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \cdot \sum(y_i - \bar{y})^2}} = \frac{2292.25}{\sqrt{2393.5 \times 2200.88}} = \frac{2292.25}{2295.17} = \mathbf{0.999}$$

This indicates a **very strong positive** linear relationship between age and systolic blood pressure.

(b) Regression coefficients:

$$b_1 = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2} = \frac{2292.25}{2393.5} = 0.958$$

$$b_0 = \bar{y} - b_1\bar{x} = 138.88 - 0.958 \times 51.25 = 89.79$$

$$\boxed{\hat{Y} = 89.79 + 0.958X}$$

(c) Prediction for age = 50:

$$\hat{Y} = 89.79 + 0.958 \times 50 = 137.7 \text{ mmHg}$$

(d) R^2:

$$R^2 = r^2 = (0.999)^2 = 0.997$$

99.7% of the variation in systolic blood pressure is explained by age. The remaining 0.3% is due to other factors not included in this model.

Example 2: Interpreting SPSS Output

A researcher uses SPSS to examine the relationship between **BMI (kg/m²)** and **fasting blood glucose (mg/dL)** in 60 adults. The following output is obtained:

Model Summary:

R	R²	Adjusted R²	Std. Error
0.612	0.375	0.364	14.23

Coefficients:

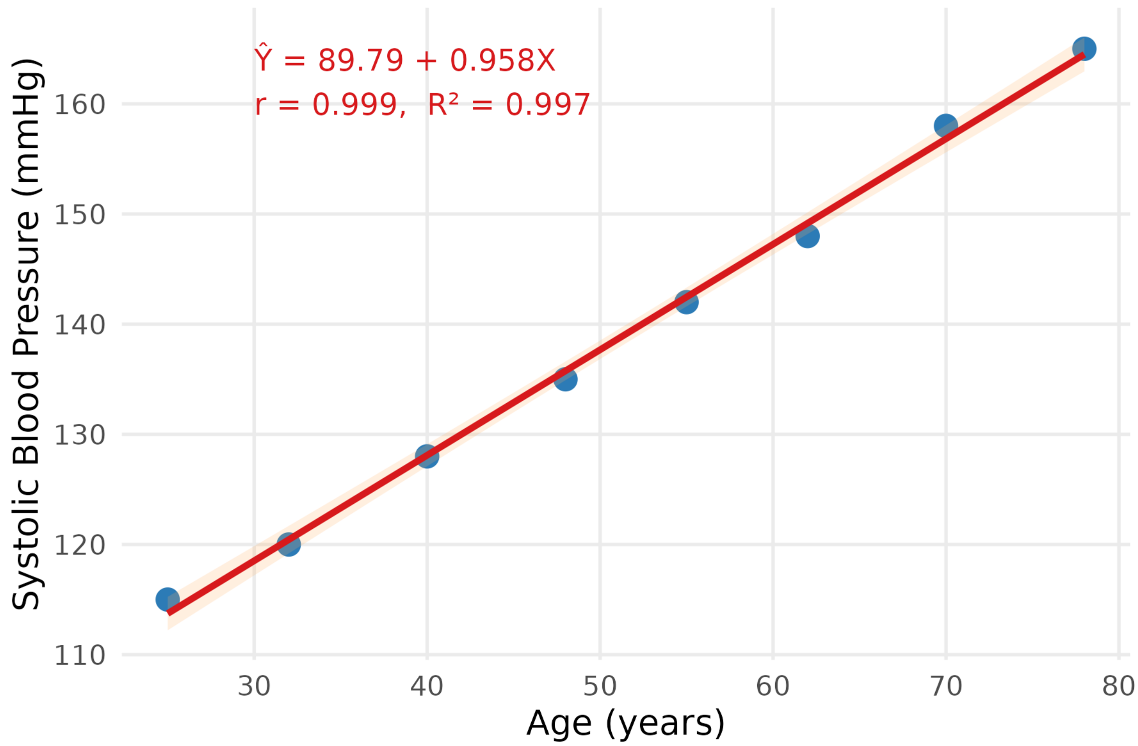

Figure 1: Scatter plot with fitted regression line: Age vs. Systolic Blood Pressure

	B	Std. Error	t	Sig.
Constant	48.32	11.45	4.22	< 0.001
BMI	2.91	0.46	6.33	< 0.001

Interpretation

Correlation: $r = 0.612$ — a **moderate-to-strong positive** relationship between BMI and fasting blood glucose.
Regression equation:

$$\hat{Y} = 48.32 + 2.91 \times \text{BMI}$$

Slope ($b_1 = 2.91$): For each 1 kg/m² increase in BMI, fasting blood glucose increases by **2.91 mg/dL** on average, holding other factors constant.
Intercept ($b_0 = 48.32$): The predicted fasting glucose when BMI = 0 is 48.32 mg/dL. *(Note: BMI of 0 is not a meaningful value in practice — the intercept is simply a mathematical anchor for the line.)*
$R^2 = 0.375$: BMI explains **37.5%** of the variation in fasting blood glucose. The remaining 62.5% is due to other factors (diet, genetics, physical activity, etc.).
Significance ($p < 0.001$): The slope is highly significant — BMI is a statistically significant predictor of fasting blood glucose.

Prediction example: A person with BMI = 28 kg/m²:

$$\hat{Y} = 48.32 + 2.91 \times 28 = 48.32 + 81.48 = 129.8 \text{ mg/dL}$$

Exercises

Exercise 1

A study records the **hours of sleep per night** (X) and **systolic blood pressure (mmHg)** (Y) for 6 adults:

Subject	Sleep (hrs)	SBP (mmHg)
1	5	145
2	6	138
3	7	130
4	7	125
5	8	120
6	9	115

(a) Draw a scatter plot and describe the relationship.
(b) Calculate Pearson's r. What does it indicate?
(c) Find the regression equation $\hat{Y} = b_0 + b_1 X$.
(d) Predict the SBP of a person who sleeps 6.5 hours per night.
(e) Calculate R^2 and interpret it.

Exercise 2

A researcher uses SPSS to study the relationship between a patient's **age (years)** and **total cholesterol (mg/dL)**. The SPSS output gives the following results:
Model Summary: R = 0.548, R^2 = 0.300
Coefficients:

	B	Std. Error	t	Sig.
Constant	120.4	18.6	6.47	< 0.001
Age	1.85	0.31	5.97	< 0.001

(a) Write the regression equation.
(b) Interpret the slope in the context of this study.
(c) What proportion of the variation in cholesterol is explained by age? Is this a strong model?
(d) Predict the total cholesterol of a 55-year-old patient.
(e) Is the predictor (age) statistically significant at $\alpha = 0.05$? How do you know?

Exercise 3 — Reflection Questions

Answer the following in 2–3 sentences each:
(a) A researcher finds that the correlation between ice cream sales and drowning deaths is $r = 0.85$. Does this mean eating ice cream causes drowning? Explain using the concept of correlation vs. causation.
(b) In a regression model predicting patient recovery time from age ($R^2 = 0.22$), a colleague says *"this model is useless because it only explains 22% of the variation."* Do you agree? What additional information would help evaluate this claim?
(c) Why is it generally not appropriate to use a regression equation to predict Y for values of X that are far outside the range of the original data?

End of Chapter 6

End of Course

Congratulations on completing all six chapters of Introduction to Biostatistics!
You have covered the full journey from understanding what data is, to summarizing it, to understanding distributions, making estimates, testing hypotheses, and finally modeling relationships between variables.
These tools form the foundation of evidence-based practice in health science.